

SOFTWARE REQUIREMENT SPECIFICATIONS PROPOSAL

ONLINE BANK SYSTEM



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ABSTRACT:

The increasing demand for convenient banking services has highlighted the limitations of traditional banking methods, which often require physical branch visits for performing different operations. This inefficiency leads to customer dissatisfaction, especially in today's fast-paced digital world. The proposed Online Bank System offers a solution by providing a user-friendly digital platform for customers to manage their bank accounts remotely. With features such as online fund transfers, transaction history tracking, and real-time balance inquiries, the system addresses the need for convenient banking. By digitizing these services, the system reduces the need for in-person visits, thereby improving customer satisfaction and operational efficiency for banks.

INTRODUCTION:

In today's fast-paced world, traditional banking methods often struggle to meet the needs of modern customers. Many people find it challenging to manage their financial activities because of the limitations of physical bank branches. Tasks that should be simple, like checking account balances, transferring money, or paying bills, can take a lot of time. Customers must visit a bank during specific hours, which can cause delays for those with busy lives or urgent financial needs. Long lines and endless paperwork only add to the frustration. For banks, keeping physical branches open and handling transactions by hand leads to higher operational costs and challenges in managing resources. Furthermore, people living in remote areas may find it hard to access banking services, making traditional banking feel even more restrictive.

The Online Bank System aims to solve these problems by offering a strong digital banking platform that meets the needs of today's customers. With this system, people can access a variety of banking services anytime, anywhere, without having to visit a physical branch. Whether it's checking account balances, sending money to friends, or viewing transaction history, customers can do it all through an easy-to-use online portal. This not only saves time but also reduces the workload for banks by automating everyday tasks. Additionally, the system prioritizes security, using measures like real-time monitoring to keep sensitive information safe from threats.

This new online banking system also improves the overall experience for users. The platform is designed to be user-friendly, allowing for easy navigation and access to personalized services. Features like customers manage their finances better by providing insights into their spending habits. By cutting down on the need for physical paperwork, the system promotes eco-friendly banking practices that align with the growing concern for the environment.

Ultimately, the Online Bank System offers a fresh solution that addresses the common frustrations of traditional banking. It provides a quicker, more efficient, and accessible way for people to handle their finances. By embracing this modern approach, banks can strengthen

customer loyalty, attract new clients, and stay competitive in a changing market. This system not only meets the demands of today's consumers but also opens the door to financial services for those who previously faced barriers, ensuring that everyone can benefit from essential banking services, regardless of their situation. The Banking System aims to solve the problems of traditional banking, which often frustrate customers. By removing the need for physical bank branches, the Online Bank allows users to access a wide range of banking services anytime and anywhere. Customers can check their account balances, transfer money, pay bills, and manage their finances with just a few clicks, all from home. The Online Bank is designed to be easy to use, giving people the freedom to control their financial activities without being limited by banking hours or long lines. With a strong focus on safety and efficiency, this innovative service makes banking easier and more enjoyable for everyone. As the first Online Bank, it aims to change the way people think about banking, making it more accessible and convenient for all, no matter where they are or how busy their lives may be. This exciting new approach is not just about technology; it's about transforming the way we interact with our finances in the modern world

PROBLEM STATEMENT:

How can we manage our Bank Accounts Online?

OBJECTIVES:

- Our main objective is to create an efficient online banking application that allows customers to perform banking activities from anywhere without visiting a physical bank branch.
- Enabling core banking functions like account management, money transfers (Deposits and Withdraws), and viewing transaction history in a user-friendly interface.
- Streamline everyday banking tasks for customers, saving time and reducing the need for manual paperwork or in-person assistance.
- Manage all admin related work like total funds, active accounts, total intakes and clearings etc. in user-friendly environment.

TOOLS AND TECHNOLOGIES:

1. Programming Languages: C#.
2. Frameworks: MICROSOFT ASP.NET.
3. UI Frameworks: MS Windows Form Application.
4. Database: SQL Server.
5. IDE: Microsoft Visual Studio.
6. Designer: Blend for VS.
7. Referencing: EndNote.
8. Diagrams: Star UML.
9. Wireframes: Pencil.
10. VCS: Git & GitHub.
11. Text Editor: MS Word.
12. User devices (computers, smartphones etc.)
13. **Mermaid (1)** (for timeline and Gantt chart)

VISION:

We want to create an online bank system so that the users can perform different banking operations without physically going to the bank.

SCOPE:

We want to create a bank system in which user can manage his/her account. User can add funds in his/her account by using JazzCash and EasyPaisa after verification from the listed bank services. User can transfer funds from his account to other accounts using unique IBAN number. User can view all the transaction details including deposits and withdrawals. On the other hand, the admin can manage the stats on higher levels, can calculate total amounts of Funds in the bank, active accounts, and can see the user information of users.

FUNCTIONAL REQUIREMENTS:

Actors:

User: A bank customer who can manage their personal account, transfer funds, view transactions and perform other operations.

Admin: A system administrator responsible for managing overall account activity and monitoring the bank system.

Users:

- The user shall be able to sign up and create a new account using their personal details.
- The user shall log in to the system using their credentials.
- The user shall be able to view their account information on the dashboard.
- The user shall be able to view their current account balance.
- The user shall be able to view recent transactions.
- The user shall be able to add money to their account through the "Add Money" option.
- The user shall be able to transfer funds to another account using the "Send Money" option.
- The user shall be able to view a history of their transactions.
- The user shall be able to access the "My Account" section to view detailed account information.

Admin:

- The admin shall log in using their admin credentials.
- The admin shall be able to view the total balance of all accounts in the system.
- The admin shall be able to view total withdrawals across all accounts.
- The admin shall be able to view total deposits across all accounts.
- The admin shall be able to view a list of all active accounts.
- The admin shall be able to view and manage user data on a dedicated page.
- The admin shall be able to generate reports based on account activity.
- The admin shall be able to monitor overall system performance.

NON-FUNCTIONAL REQUIREMENTS:

Response Time: The system must answer the query in less than two seconds.

Capacity: The system must have storage to store the data of 1000 users.

Throughput: The system must handle the 10 users per second.

Security: All sensitive user data (e.g., passwords, transaction history) must be secure in the database.

Scalability: The system must be scalable to handle a growing number of users without affecting performance.

Performance: The system must load the dashboard within 2 seconds for a smooth user experience.

Reliability: The system must have 99.9% uptime.

Usability: The system must be designed user friendly to ease non-technical users as well.

Availability: The system must be available 24/7 to ensure users can access their accounts and perform transactions at any time.

Maintainability: The system must support regular updates and patches without causing any downtime or disruption in service.

Recoverability: In case of system failure, the system must recover within 2 hours to minimize service interruption.

Serviceability: The system should diagnose and report any issues or failures within 30 minutes to allow prompt troubleshooting.

Data Integrity: The system must maintain 99.99% accuracy in handling transaction data to ensure the correctness of all deposits and withdrawals.

Regulatory Compliance: The system must comply with all financial and data privacy regulations, ensuring the protection of user information.

Interoperability: The system must integrate smoothly with external systems like payment gateways and banking APIs, ensuring seamless transactions.

Utilization: The system must not exceed 75% of CPU utilization during peak hours to maintain performance stability.

Environmental: The system should operate efficiently in a temperature range between 10°C and 45°C, ensuring hardware and software reliability.

BUSINESS REQUIREMENTS:

Cost Efficiency: The system must reduce the cost of customer service by automating routine banking operations

Data Management: The system must allow admins to generate reports on financial data, such as deposits, withdrawals, and account activity.

User Accessibility: The system must be accessible 24/7, allowing users to perform banking tasks at their convenience.

Operational Efficiency: The system must improve overall bank efficiency by automating repetitive tasks such as processing transactions, account updates, and balance inquiries, thereby reducing the need for manual intervention and minimizing operational costs.

Profit Generation: The system must help increase profits by offering banking services, such as faster fund transfers, higher withdrawal limits, and exclusive features account holders, which will generate additional revenue for the bank.

PHYSICAL REQUIREMENTS:

Personal Computers/Laptops: Must have at least 4GB of RAM, a dual-core processor, and a stable internet connection (preferably 5 Mbps or higher). The system shall be compatible with Windows 10, macOS, or Linux operating systems.

Smartphones/Tablets: Mobile devices running iOS 12.0+ or Android 8.0+ with at least 2GB of RAM are required for seamless access. Users must have a 4G or better network connection for optimal performance on mobile platforms.

Web Browser Requirements: Users shall access the system via modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari. The browser must support HTML5, CSS3, and JavaScript to ensure full compatibility with the system's features.

Backup Device: Physical backup devices such as external hard drives with a minimum storage capacity of 2TB, connected to the main database server to perform daily backups.

WIREFRAMES:

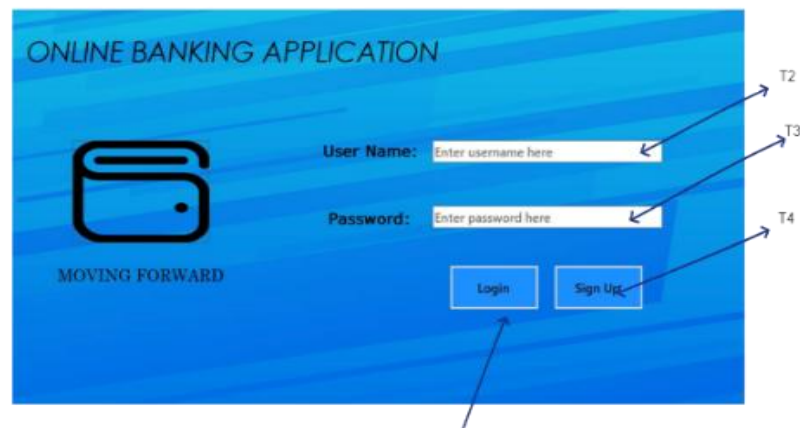


Figure 1: Login Page

REQUIREMENTS	USER	ADMIN
T1	As a user I shall login into the system and redirected to user dashboard when I click this button.	As an admin I shall login into the system and jump to admin panel when I click this button.

T2	As a user I shall enter my unique username i.e. IBAN number in this field to login into my account.	As an admin I shall enter my unique username to login into admin account.
T3	As a user I shall enter my password to login into my account.	As an admin I shall enter my password to login into admin panel.
T4	As a user I shall move to signup page to create new user account when I click this button.	

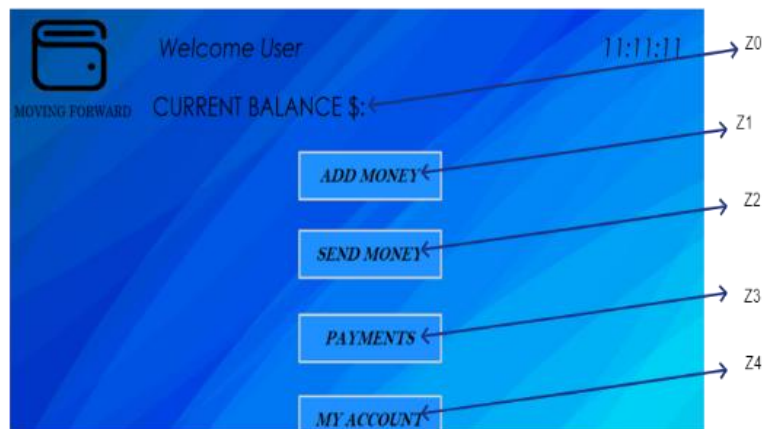


Figure 2: User Dashboard

REQUIREMENTS	USER
Z0	As a user I shall see the current balance in my account from this.
Z1	As a user, I shall be redirected to Add Money page, to add funds in my account when I click this button.
Z2	As a user, I shall jump to Send Money page to send money from my account to other ones when I click this button.
Z3	As a user, I shall move to Payments page to check transaction history when I click this button.
Z4	As a user, I shall be redirected to My Account page to see my details when I click this button.

The screenshot shows a blue-themed 'ADD MONEY' interface. At the top left is a wallet icon and the text 'MOVING FORWARD'. The title 'ADD MONEY' is at the top right. Below are three input fields: 'IBAN NUMBER:' with the value '1111 111 11111', 'AMOUNT:' with the value '\$\$\$', and 'ENTER OTP PIN:' with the value '1234'. Each field has a left-pointing arrow. At the bottom is a blue button labeled 'CONFIRM TRANSACTION'. Four arrows on the right point to these elements: M1 points to the IBAN field, M2 to the Amount field, M3 to the OTP PIN field, and M4 to the Confirm Transaction button.

Figure 3: Add Money Page

REQUIREMENTS	USER
M1	As a user, I shall enter my unique IBAN number in this field to add funds in my account.
M2	As a user, I shall enter the amount to receive into my account from other account in this field.
M3	As a user, I shall enter the OTP i.e. one time password in this field for verification.
M4	- As a user, I shall confirm transaction in my account when I click this button. - As a user, I shall be redirected to user dashboard after clicking this button.

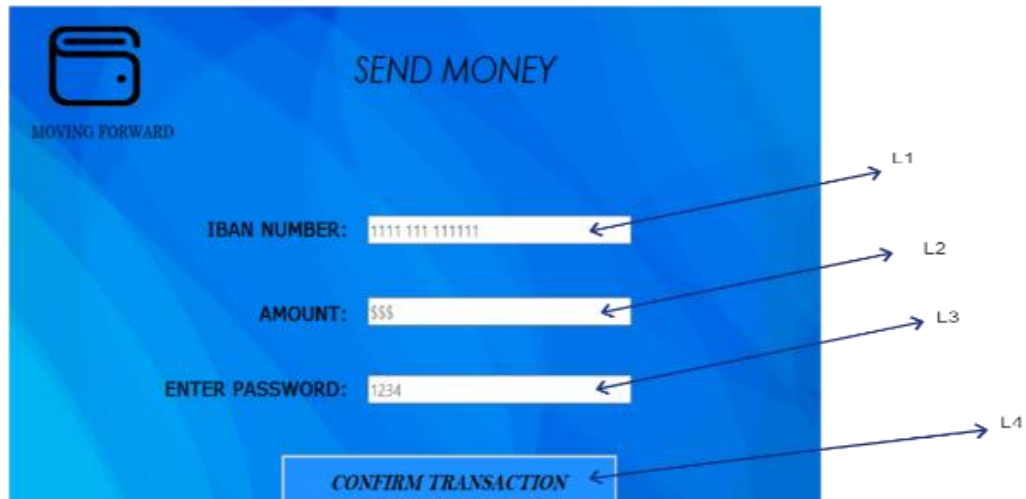


Figure 4: Send Money Page

REQUIREMENTS	USER
L1	As a user, I shall enter recipient's unique IBAN number in this field to add funds in his account.
L2	As a user, I shall enter the amount to send from my account to other account in this field.
L3	As a user, I shall enter my password in this field for verification.
L4	- As a user, I shall confirm transaction i.e. send money from my account when I click this button. - As a user, I shall be redirected to user dashboard after clicking this button.

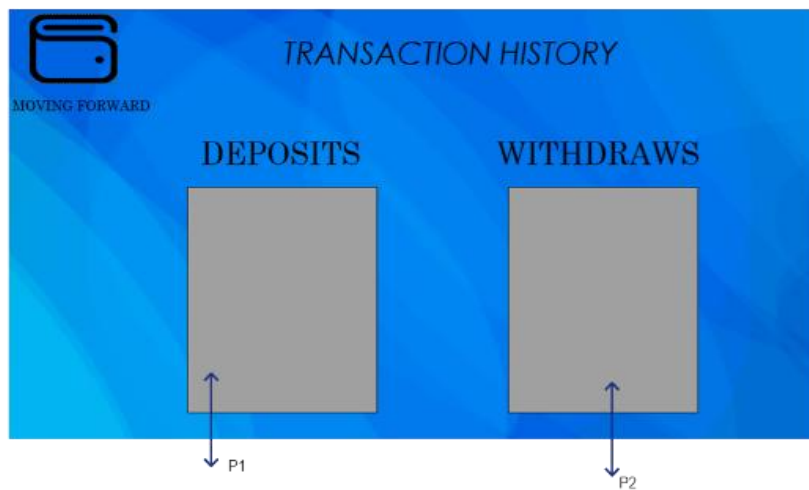
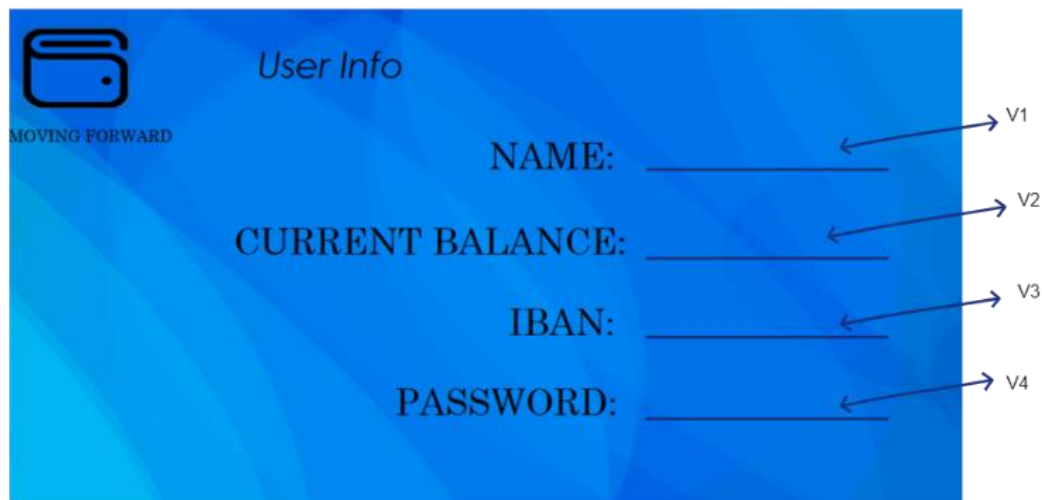


Figure 5: Transaction History Page

REQUIREMENTS	USER
P1	As a user, I shall view details of deposit history of funds in my account by using this data table.
P2	As a user, I shall view details of withdrawal history of funds from my account to other accounts by using this data table.



The image shows a 'User Info' form on a blue background. In the top left corner, there is a logo of a document with a checkmark and the text 'MOVING FORWARD'. The form contains four input fields, each with a label and a verification label (V1-V4) indicated by an arrow pointing to the input field:

- NAME:** [input field] (labeled V1)
- CURRENT BALANCE:** [input field] (labeled V2)
- IBAN:** [input field] (labeled V3)
- PASSWORD:** [input field] (labeled V4)

Figure 6: User Details Page

REQUIREMENTS	USER
V1	As a user, I shall view my name that I entered while creating my account
V2	As a user, I shall see the current balance in my account from this.
V3	As a user, I shall see the my unique IBAN number that has assigned to me, from this.
V4	As a user, I shall view my current password that has been stored in the system's database using this.



Figure 7:Admin Dashboard

REQUIREMENTS	ADMIN
K1	As an Admin, I shall view the total funds in the bank, this includes the current balance of all the active users.
K2	As an Admin, I shall view the total withdrawals made by the active users of the bank, with this.
K3	As an Admin, I shall view the total deposits made by the active users of the bank in their accounts, with this.
K4	As an Admin, I shall see the number of active users of the bank using this.
K5	As an Admin, I shall be redirected to User Details page, to see user details, by clicking this button.

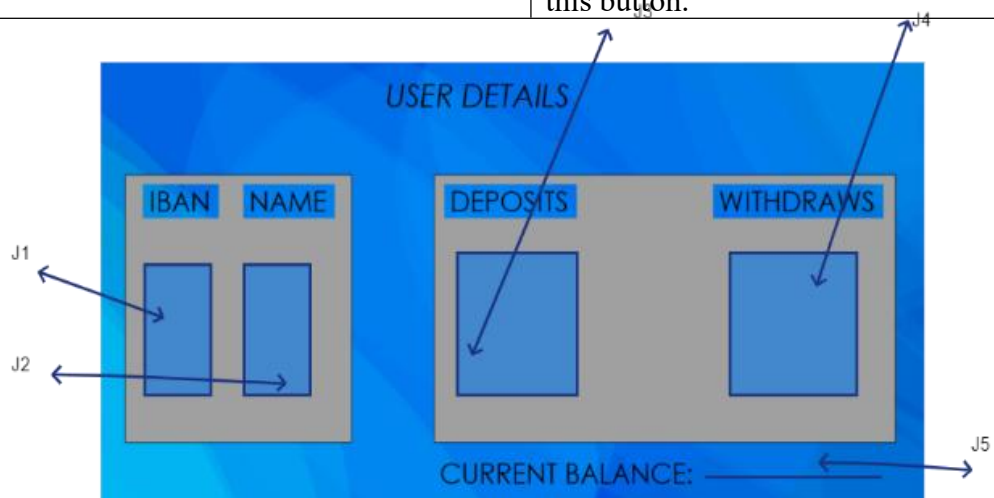


Figure 8: User Details Page

REQUIREMENTS	ADMIN
J1	- As an Admin, I shall view the unique IBAN number of all the active users of the bank. - As an Admin, I shall view transaction details of selected user by clicking this. - As an Admin, I shall view the current balance of the selected user by clicking this.
J2	- As an Admin, I shall view the name of all the active users of the bank. - As an Admin, I shall view transaction details of selected user by clicking this. - As an Admin, I shall view the current balance of the selected user by clicking this.
J3	As an Admin, I shall view deposits made by the selected user in this data table.
J4	As an Admin, I shall see withdrawals made by the selected user in this data table.
J5	As an Admin, I shall see the current balance of the selected user

SIGN UP

First Name: → A1

Last Name: → A2

Email: → A3

Date of Birth: → A4

Contact: → A5

Password: → A6

Confirm Password: → A7

→ A8

Figure 9: Sign Up Page

REQUIREMENTS	User
A1	As a user, I shall enter my first name to create my account using this field.
A2	As a user, I shall enter my last name to create my account using this field.
A3	As a user, I shall enter my email address so that I receive emails regarding my bank account.
A4	As a user, I shall enter my Date of birth to create my account using this field.
A5	As a user, I shall enter my contact number so that bank can contact me, using this field.
A6	As a user, I shall enter my password here so that whenever I log into my account password could be used as verification.
A7	As a user, I shall re enter my password in this field so that any kind of ambiguity can be resolved.
A8	- As a user, I shall create my new account by providing my details, when I click this button. - As a user, I shall be redirected to login page after my account has been created when I click this button.

USER STORIES:

Users:

- As a user, I want to sign up using my personal details so that I can create my account and use the banking application.
- As a user, I want to log in using my credentials so that I can securely access my account.
- As a user, I want to view my account information on the dashboard so that I can see an overview of my account details at a glance.
- As a user, I want to view my current account balance so that I can keep track of how much money I have.
- As a user, I want to view recent transactions so that I can monitor my spending and verify completed transactions.
- As a user, I want to add money to my account through the "Add Money" option so that I can increase my account balance easily.
- As a user, I want to transfer funds to another account using the "Send Money" option so that I can make payments or send money to others.

- As a user, I want to view a history of my transactions so that I can track my financial activities over time.
- As a user, I want to access the "My Account" section to view detailed information about my account so that I can review all my personal and financial data.

Admin:

- As an admin, I want to log in using my admin credentials so that I can securely access the admin panel and manage the system.
- As an admin, I want to view the total balance of all accounts so that I can monitor the overall financial standing of the bank.
- As an admin, I want to view the total withdrawals across all accounts so that I can track the money being taken out of the system.
- As an admin, I want to view the total deposits across all accounts so that I can monitor the money being added to the system.
- As an admin, I want to view a list of all active accounts so that I can track current users and their activity status.
- As an admin, I want to view and manage users data on a dedicated page so that I can monitor specific user information as needed.
- As an admin, I want to monitor overall system performance so that I can ensure the system runs smoothly and address any technical issues promptly.
- As an admin, I want to generate reports based on account activity so that I can analyze financial trends and make informed decisions.

STORY BOARD:

1.1 User can jump to signup page by clicking signup button to create a new account.



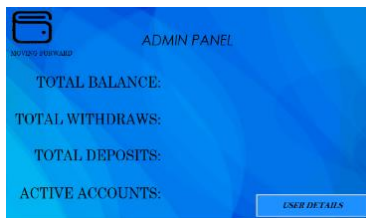
A sign-up form with fields for First Name, Last Name, Email, Date of Birth, Contact, Password, and Confirm Password. A 'Sign Up' button is at the bottom.

1.2 user can make his/her new account if already not existing after providing details and jumps to login page.



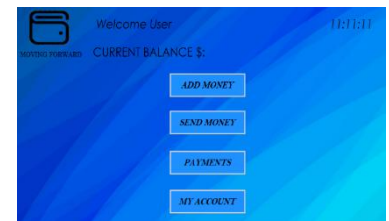
A login page titled 'ONLINE BANKING APPLICATION' with fields for User Name and Password, and 'Login' and 'Sign Up' buttons.

2. User can jump to user panel by clicking on login button after



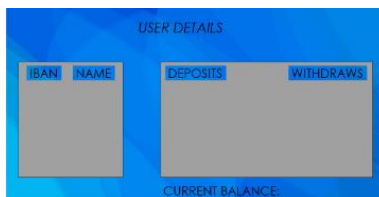
An admin panel showing 'TOTAL BALANCE', 'TOTAL WITHDRAWS', 'TOTAL DEPOSITS', 'ACTIVE ACCOUNTS', and a 'USER DETAILS' button.

3. Admin can jump to admin panel to see general details by clicking on login button after providing credentials.



A user panel showing 'Welcome User', 'CURRENT BALANCE \$', and buttons for 'ADD MONEY', 'SEND MONEY', 'PAYMENTS', and 'MY ACCOUNT'.

2.1 User can jump to add money page to add funds in his/her account by clicking add money button



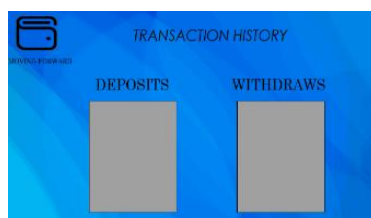
A user details page showing 'IBAN', 'NAME', 'DEPOSITS', 'WITHDRAWS', and 'CURRENT BALANCE'.

3.1 Admin can jump to user details to see all details of users, after clicking user details button



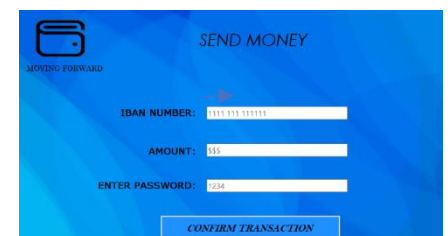
An add money form with fields for IBAN NUMBER, AMOUNT, and ENTER OTP PIN, and a 'CONFIRM TRANSACTION' button.

2.2 User can jump to send money page to send funds from his/her account to another account by clicking send money button



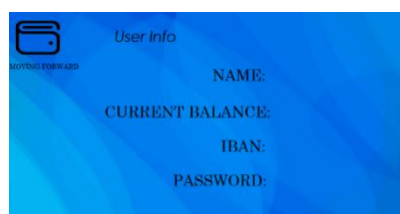
A transaction history page showing 'DEPOSITS' and 'WITHDRAWS'.

2.4 User can jump to payments page to see his/her transaction history by payments button.



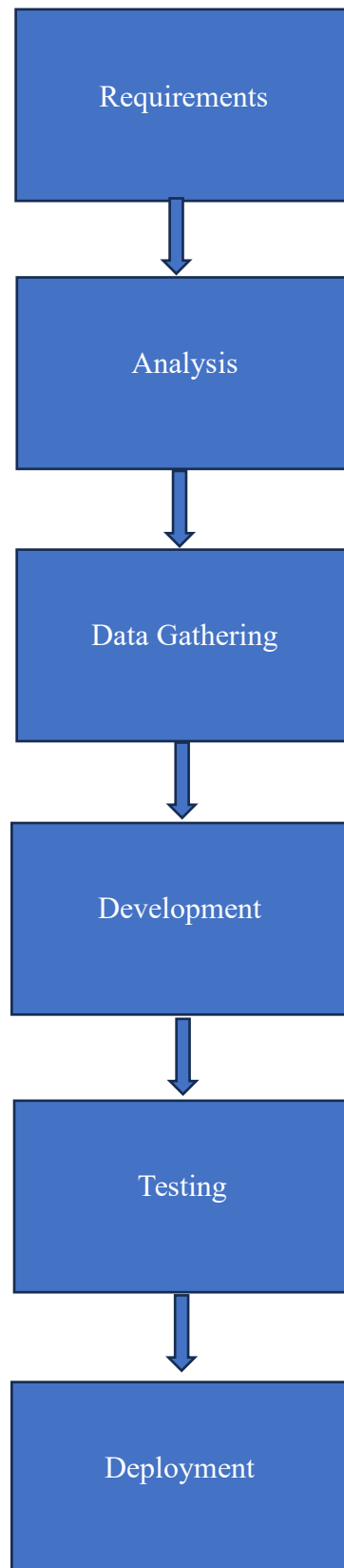
A send money form with fields for IBAN NUMBER, AMOUNT, and ENTER PASSWORD, and a 'CONFIRM TRANSACTION' button.

2.3 User can jump to my account page to see his/her information by clicking my account button.



A user info page showing 'NAME', 'CURRENT BALANCE', 'IBAN', and 'PASSWORD'.

METHODOLOGY:



Requirements:

We gathered requirements (functional and non-functional) for the Bank System, focusing on key features like account management, transactions, and the admin dashboard. We detailed specific user actions such as viewing account information and transaction history, while addressing performance to ensure a reliable banking experience.

Analysis:

We analyzed if the project was practical based on the requirements and is feasible. This involved looking at the tools and setup needed, like databases for managing user accounts and transactions.

Data Gathering:

We collected information from various sources, including user feedback to understand the needs and expectations for the Online Bank System. We also gathered information from already existing systems.

Development:

The development phase involved coding the core functionalities of the system. Using C#, SQL Server, and WinForms, we implemented the user login, registration, dashboard, and transaction management and other features

Testing:

Once development was complete, testing was conducted to ensure the system operates smoothly. We performed unit testing for individual modules like login, transaction management, and account data handling.

Deployment:

The final stage was deploying the system in a live environment. The Online Bank System was created and converted into executable file (.exe). The deployment also involved setting up a user-friendly installation process for the admin and user interfaces.

RELATED WORK:

In developing the proposal for the Online Bank System, I reviewed several existing digital banking platforms, including **SadaPay, Konnect, EasyPaisha, and JazzCash**, to understand current trends and best practices in online banking systems. These applications provided valuable insights into important banking features such as secure login systems, fund transfers, and transaction history management. The analysis of these systems helped in identifying the key requirements and functionalities that should be included in the proposed system to make it secure, efficient, and user-friendly.

During the study of these applications, several common issues were identified. Many existing banking apps have complex interfaces that may be difficult for new users to understand. Therefore, the proposed system aims to provide a simple and intuitive interface that will allow users to navigate the platform easily. In addition, some applications occasionally experience

crashes or performance-related issues, which can affect the user experience. To address this, the proposed system emphasizes stability and smooth performance. Furthermore, platforms such as **JazzCash** may sometimes experience delays in money transfers. The proposed system is intended to support faster transaction processing and near real-time fund transfers to improve efficiency and user satisfaction. By addressing these issues, the proposed Online Bank System aims to provide a more reliable, efficient, and user-friendly banking solution.

User interface of each of the above mentioned banking apps is given below.

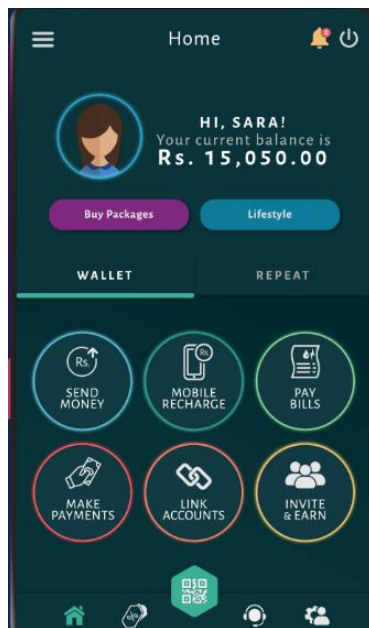


Figure 10:HBL KONNECT APP

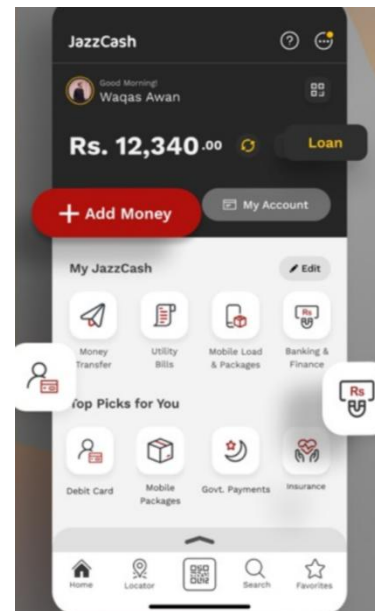


Figure 11: JAZZ CASH APP

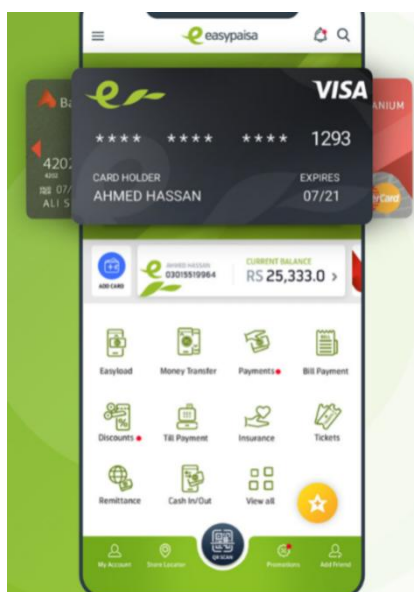


Figure 13:EASY PAISA MOBILE APP

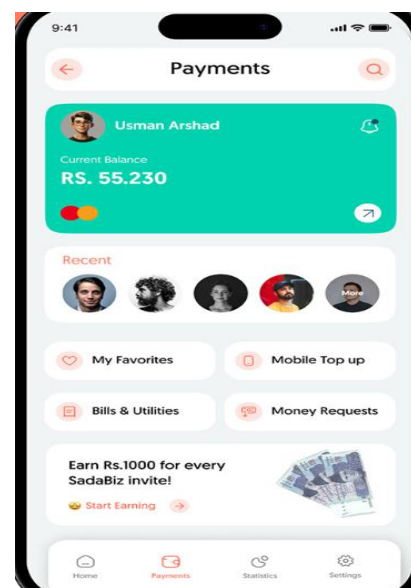


Figure 12:SADA PAY APP

With a focus on simplicity, the proposed Online Bank System aims to provide an intuitive interface that will allow users to easily manage their accounts, transfer funds, view transaction histories, and perform other essential banking tasks. The proposed system will include advanced security measures to ensure that sensitive user data, such as passwords and financial information, remains encrypted and protected. In addition, the system is intended to deliver efficient performance with fast response times and smooth transaction processing.

The proposed system will also focus on reliability and stability to minimize downtime and provide users with uninterrupted access to banking services whenever needed. Furthermore, the system is planned to be scalable so that it can support future growth and an increasing number of users while maintaining good performance. Overall, the proposed Online Bank System aims to provide a secure, user-friendly, and efficient solution for modern digital banking needs.

Individual Contribution:

As this is an individual project, all phase-wise tasks from core requirement gathering to system documentation are performed solely by the lead developer.

Laiba Nasir (Roll No: 534):

Requirement Engineering: Extracted and documented functional, non-functional, physical, and business requirements.

UI/UX & Storyboarding: Designed the user and admin panel wireframes and complete storyboards.

Agile Modeling: Developed the complete set of User Stories and mapped out the operational logic.

System Architecture & Diagrams: Designed the core Software Engineering models including the Class Diagram, Activity Diagrams, and Sequence Diagrams.

Quality Assurance: Drafted and verified all Integration and System Test Cases for critical banking modules (Login, Sign-Up, Add/Send Money, History).

Timeline/Gantt Chart:

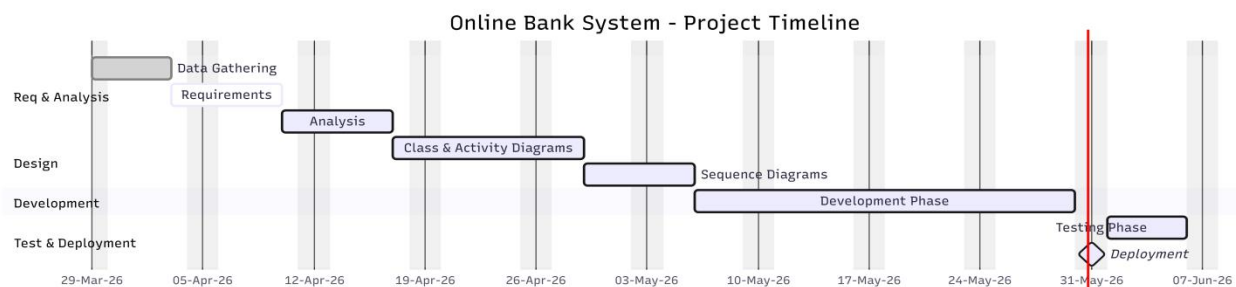


Figure 14: Timeline

Milestone	Start Date	End Date	Deliverable
Requirements Gathering	29/03/2026	05/04/2026	Software Requirements Specification (SRS) Document
System Analysis	06/04/2026	12/04/2026	Requirements Analysis Report
System Design	13/04/2026	26/04/2026	UML Diagrams (Use Case, Class & Activity Diagrams)
Database Design	27/04/2026	03/05/2026	ER Diagram and Database Schema Devsign
Interface Design	04/05/2026	10/05/2026	Website Wireframes and UI Mockups
Development Planning	11/05/2026	31/05/2026	Proposed System Architecture and Development Plan
Testing Strategy	01/06/2026	04/06/2026	Test Plan and Test Case Documentation
Final Proposal Submission	05/06/2026	07/06/2026	Complete Project Proposal Report
Final Proposal Submission	05/06/2026	07/06/2026	Complete Project Proposal Report

Figure 15:Timeline(2)

DATA GATHERING APPROACH:

To prepare the proposal for the Online Bank System, data was gathered from various sources to better understand user requirements and existing banking solutions. Feedback and observations from potential users were considered to identify common challenges faced in existing banking systems, such as long wait times, limited accessibility, and usability issues. This analysis helped in understanding the key features and improvements needed in the proposed system.

In addition, existing digital banking applications such as **SadaPay** and **Konnect** were studied to gain insight into modern banking features, user-friendly interfaces, and system functionality. Online resources, including UI/UX examples and screenshots of banking applications, were also reviewed to help guide the proposed system's interface and overall design approach.

Furthermore, personal analysis and ideas were incorporated to identify ways of improving the user experience by proposing more efficient features and better functionality. These combined sources of information helped in shaping the proposed Online Bank System to ensure that it would provide a smooth, secure, and efficient banking experience for users.

REFERENCES:

1. Technologies M. Mermaid Live Editor [Gantt Chart Online Editor]. Available from: <https://mermaid.live/>.
2. SadaPay. SadaPay Karachi,Pakistan: SadaPay; 2019 [Online Banking App]. Available from: <https://www.sadapay.pk/>.
3. Limited HB. HBL KONNECT APP Lahore,Pakistan.: HBL; 1989 [Online Banking Application.]. Available from: www.hblkonnect.com.pk/home.
4. EasyPaisha. EasyPaisha Online Paying App Islamabad, Pakistan: 2010; 2010 [Online Banking Application]. Available from: www.easypaisha.com.pk/user.
5. Limited JMFP. JazzCash Online Application Lahore,Pakistan: 2009; 2009 [JazzCash Online Banking MicroFinance app.]. Available from: www.jazzcash.com.pk/home/user.

DESIGN:

The design document of the proposed system is given below.

Class Diagram:

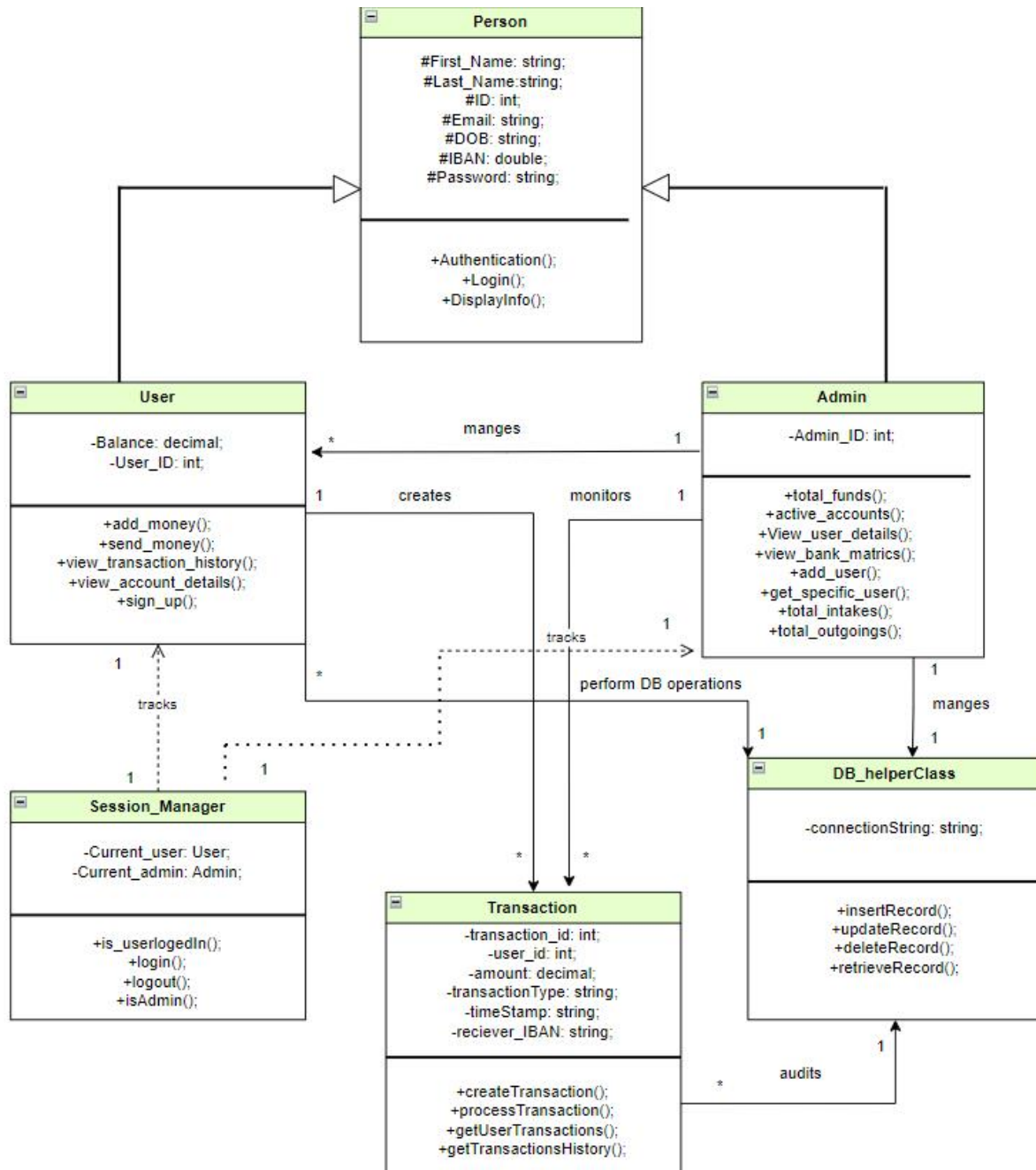


Figure 16: Class Diagram

Activity Diagrams:

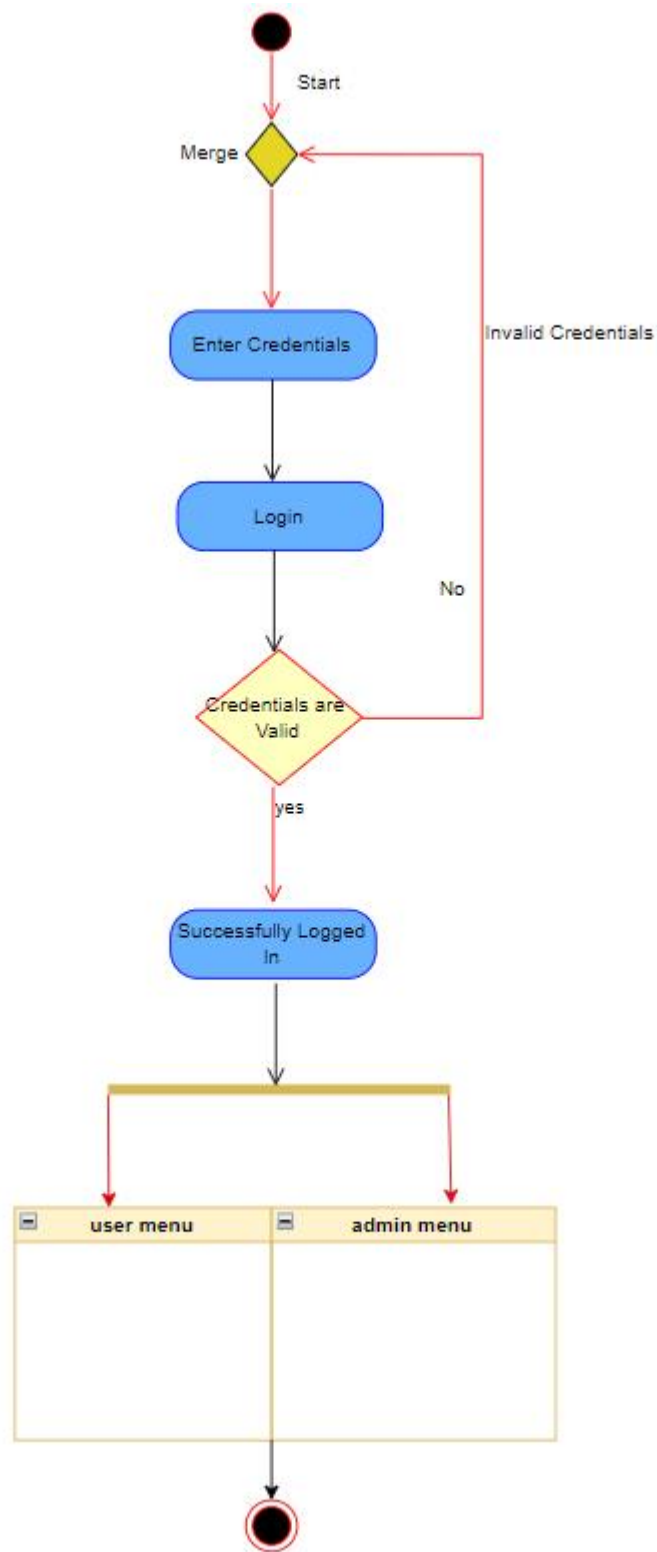


Figure 17: Login Activity Diagram

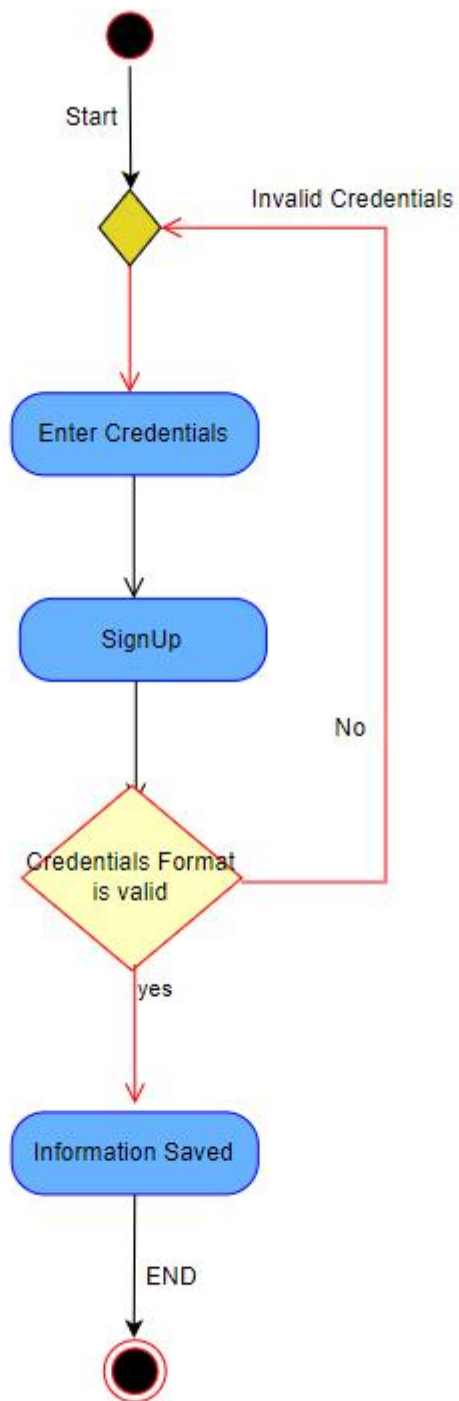


Figure 18: Sign Up Activity Diagram

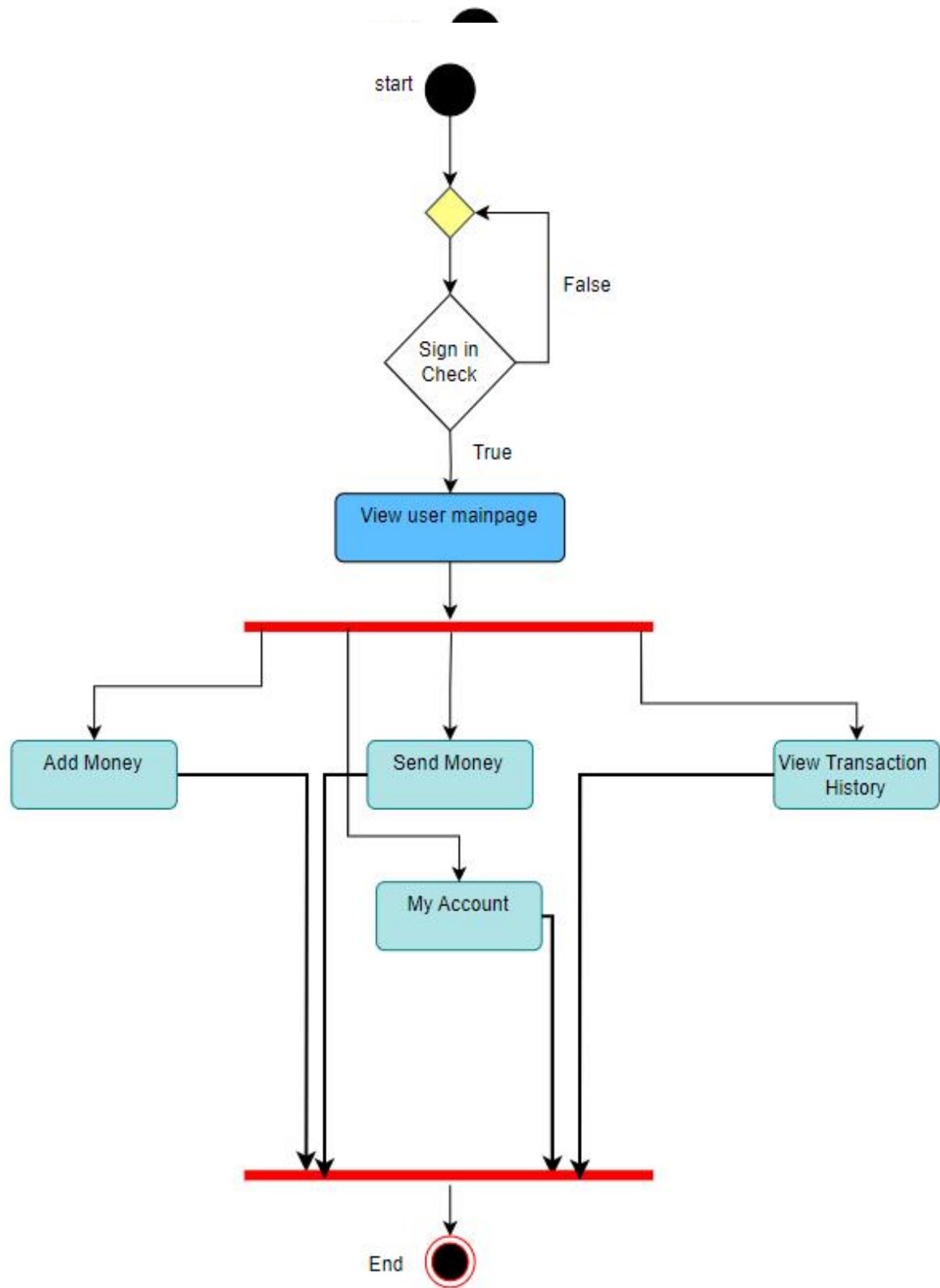


Figure 19: User Dashboard Activity Diagram

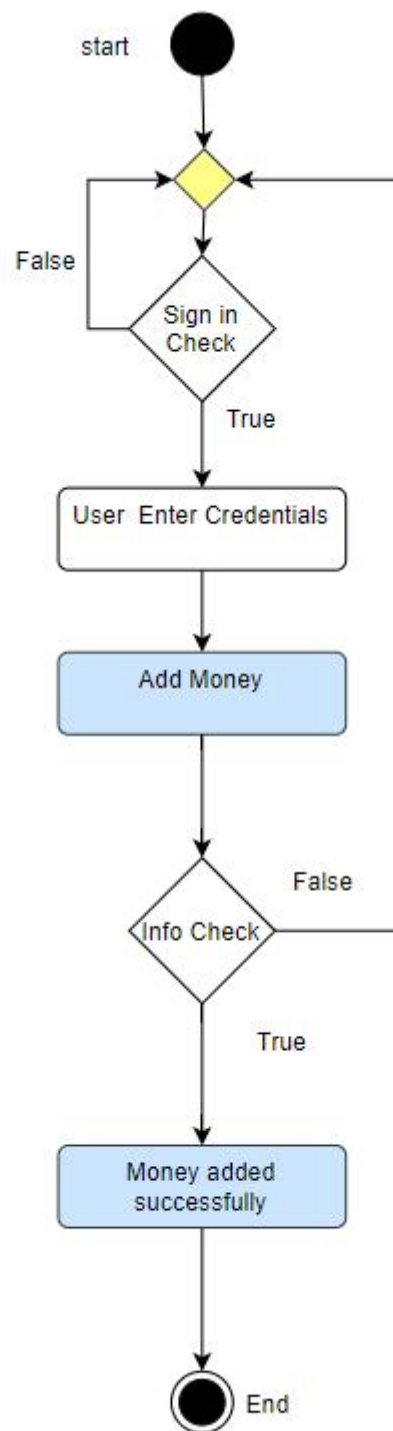


Figure 20: User Add Money Activity Diagram

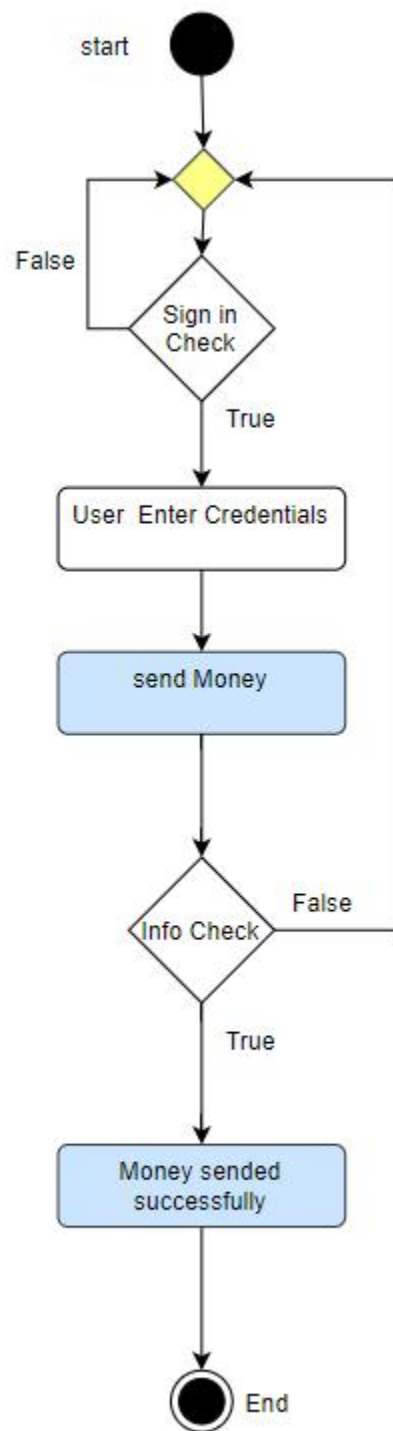


Figure 21: User Send Money Activity Diagram

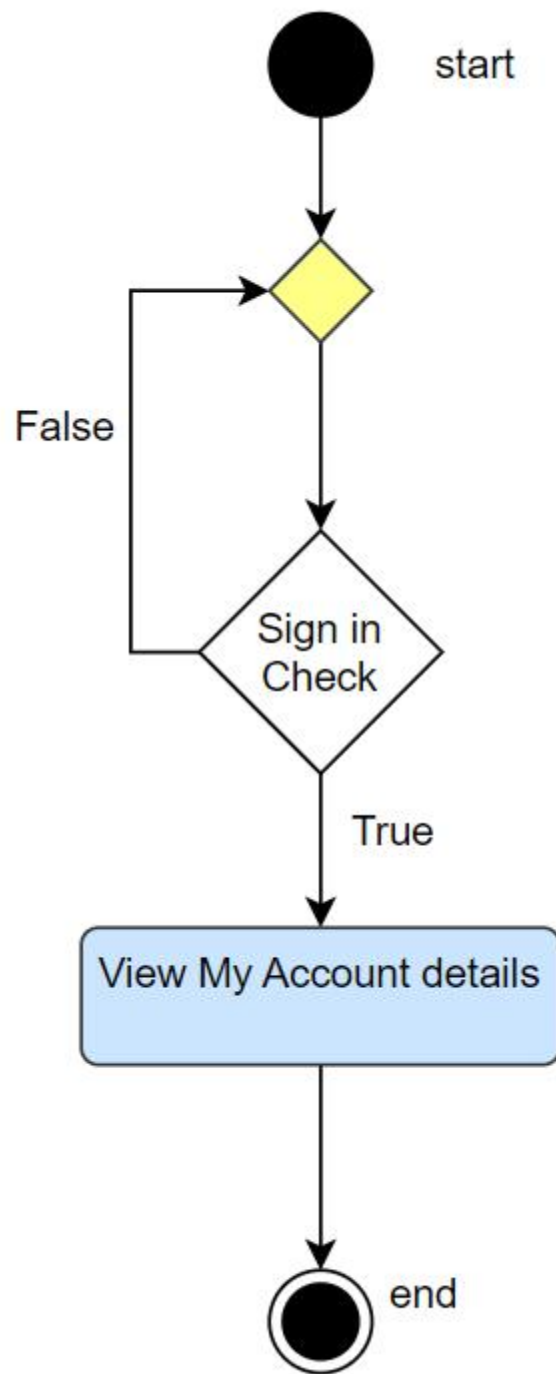


Figure 228: User View My Account Details Activity Diagram

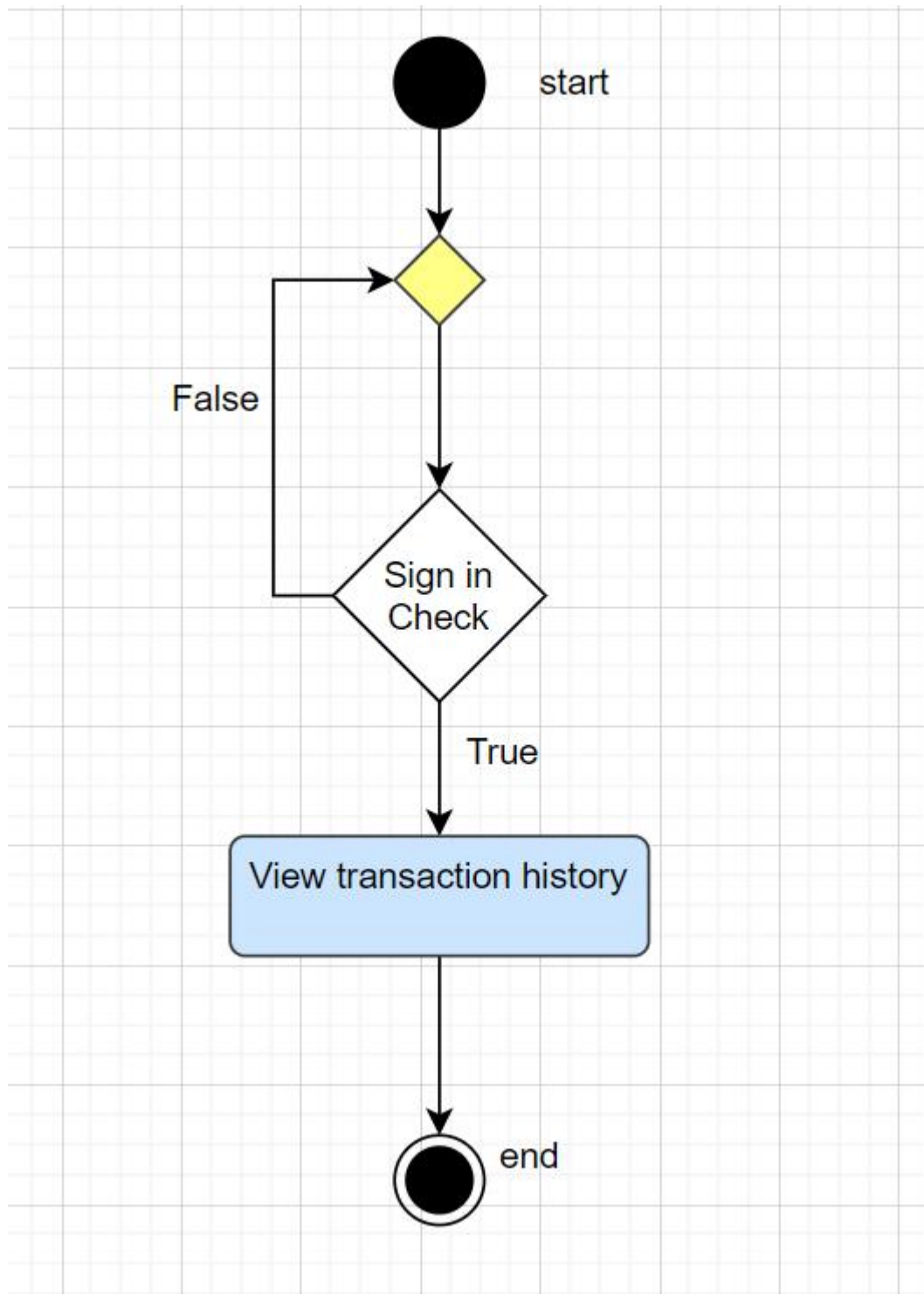


Figure 23: User View History

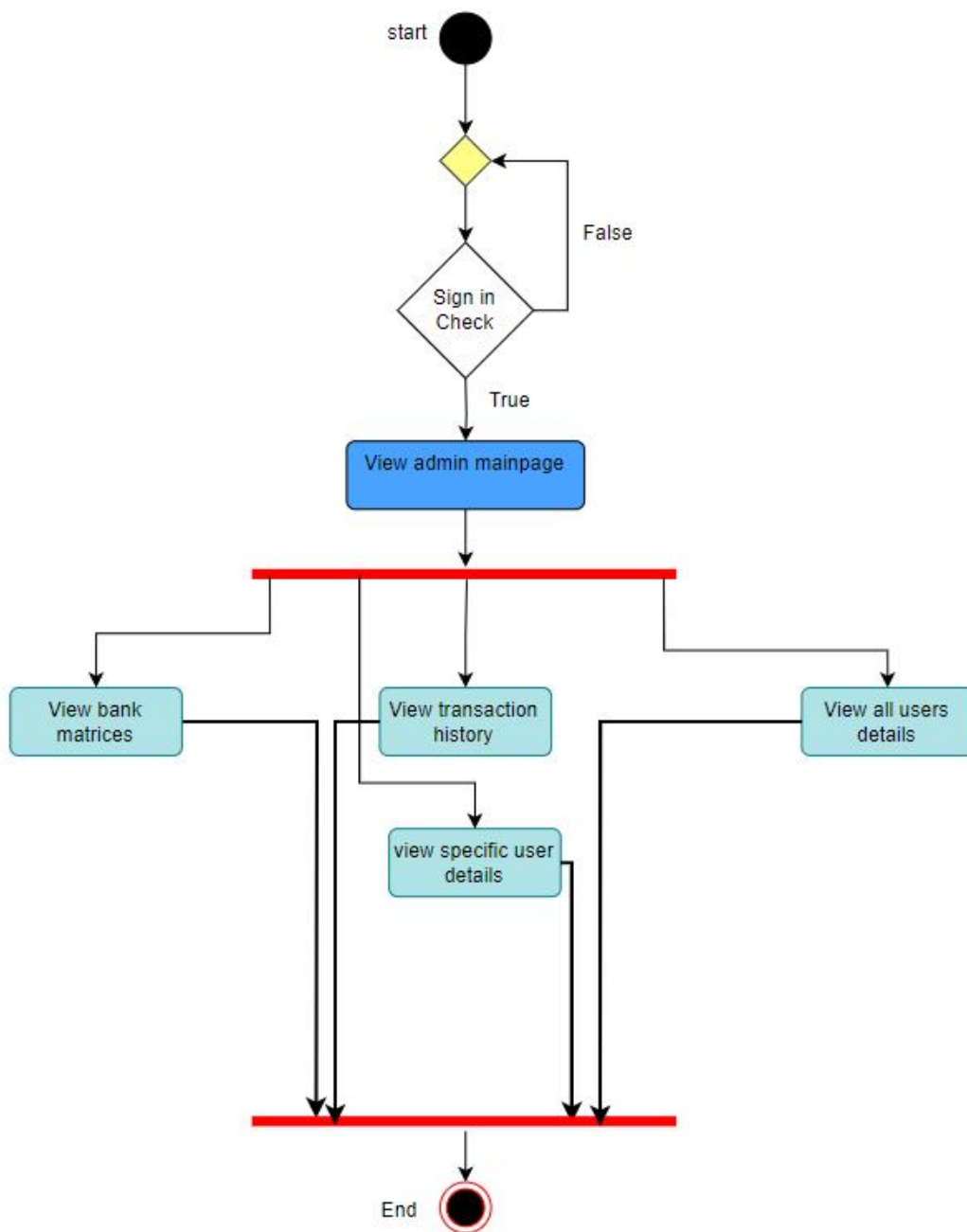


Figure 24:Admin Panel Activity Diagram

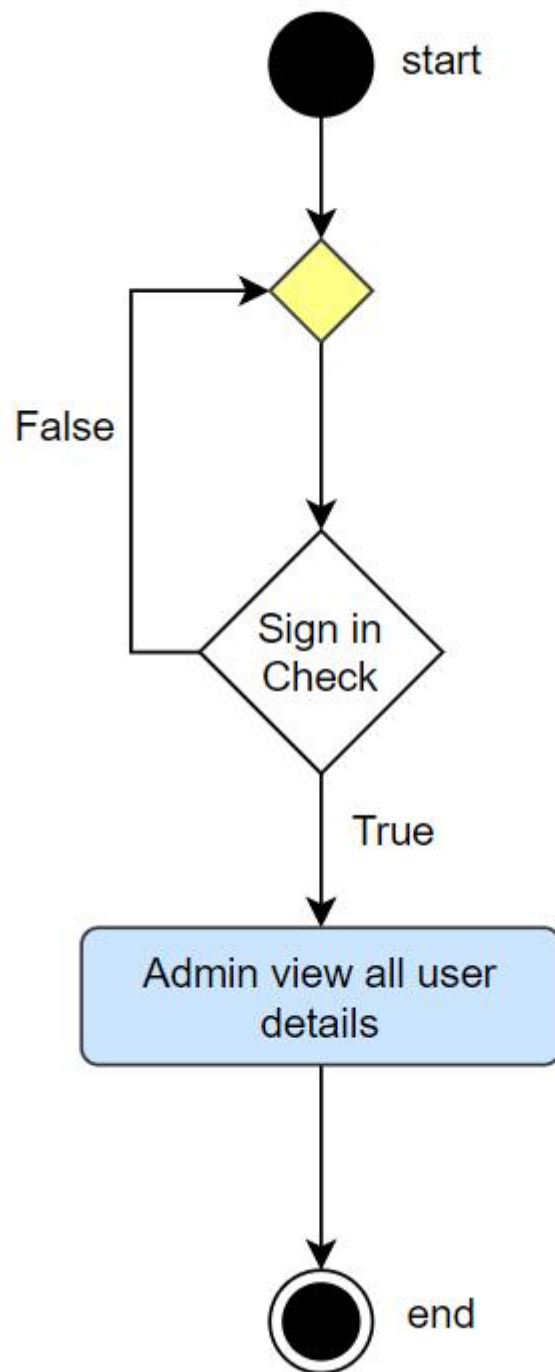


Figure 25: Admin View All User Details

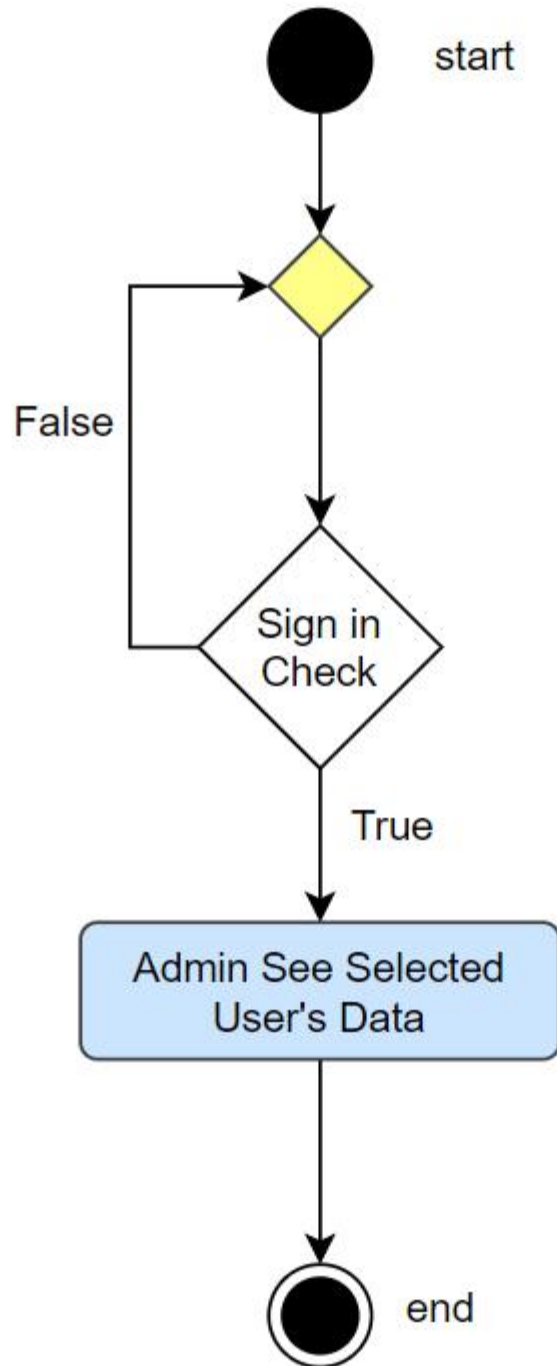


Figure 26: View Selected User Info

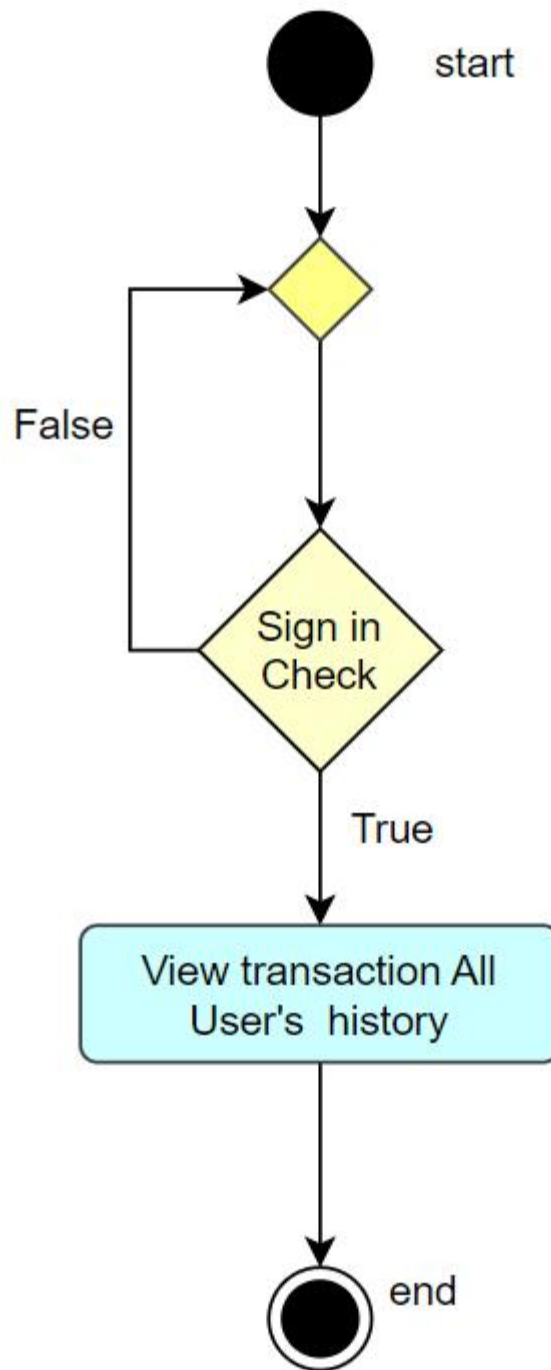


Figure 27: View all User History Activity Diagram

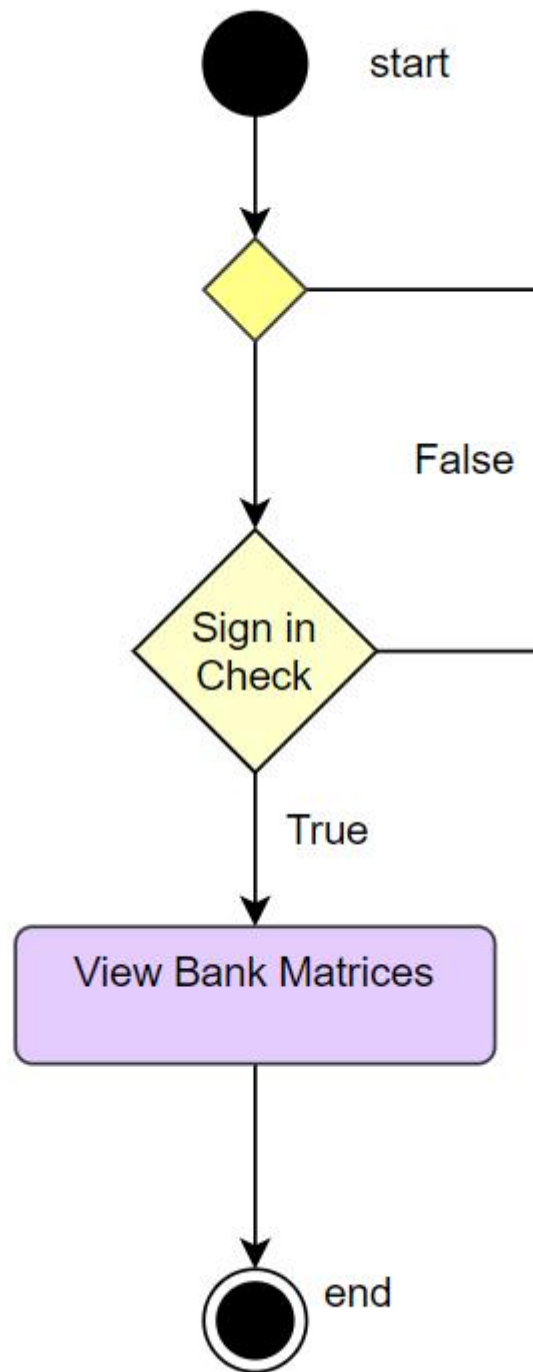


Figure 28: View Bank Matrices

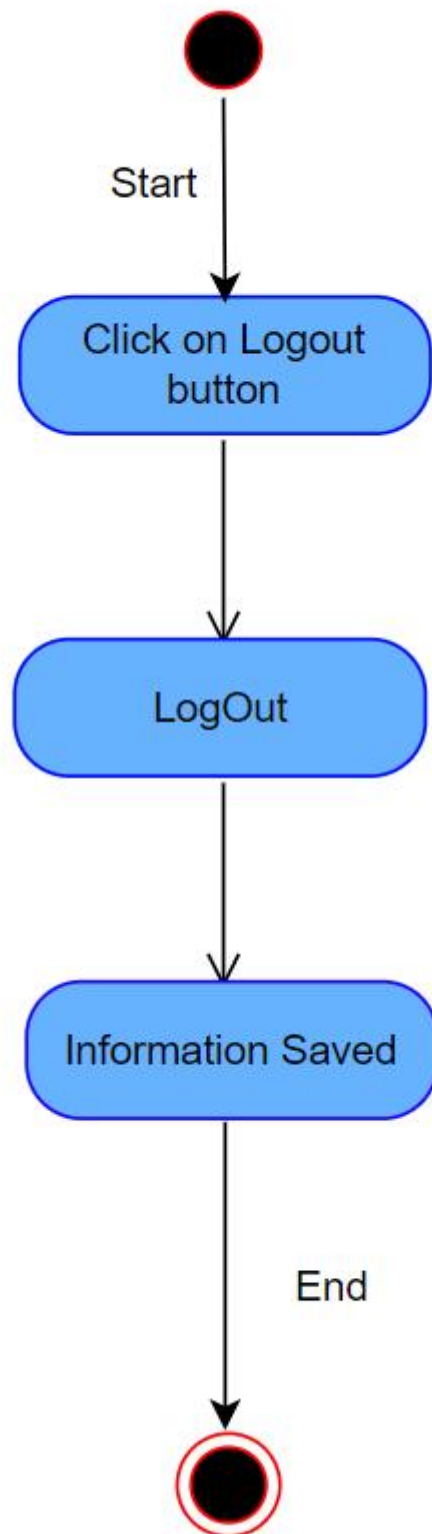


Figure 29: Log Out Activity Diagram

Sequence Diagrams:

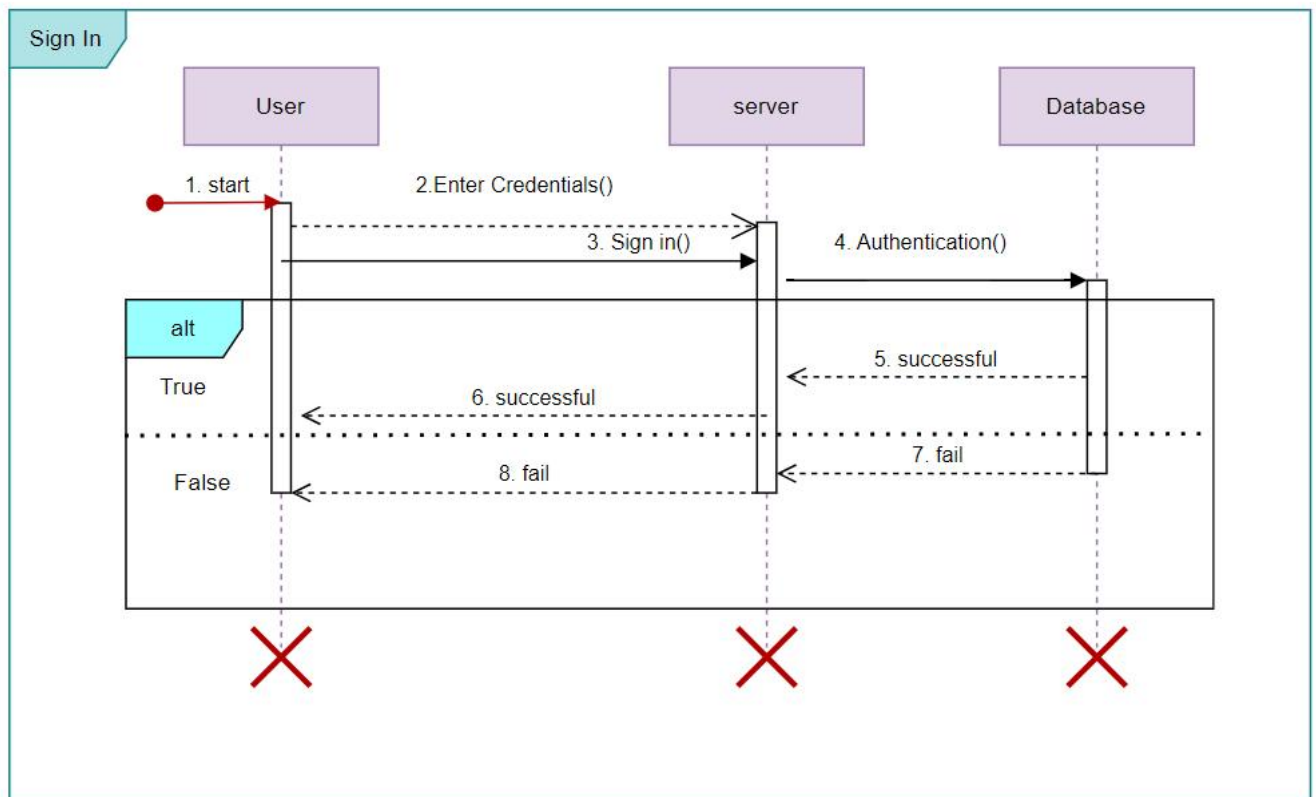


Figure 30: Login Sequence Diagram

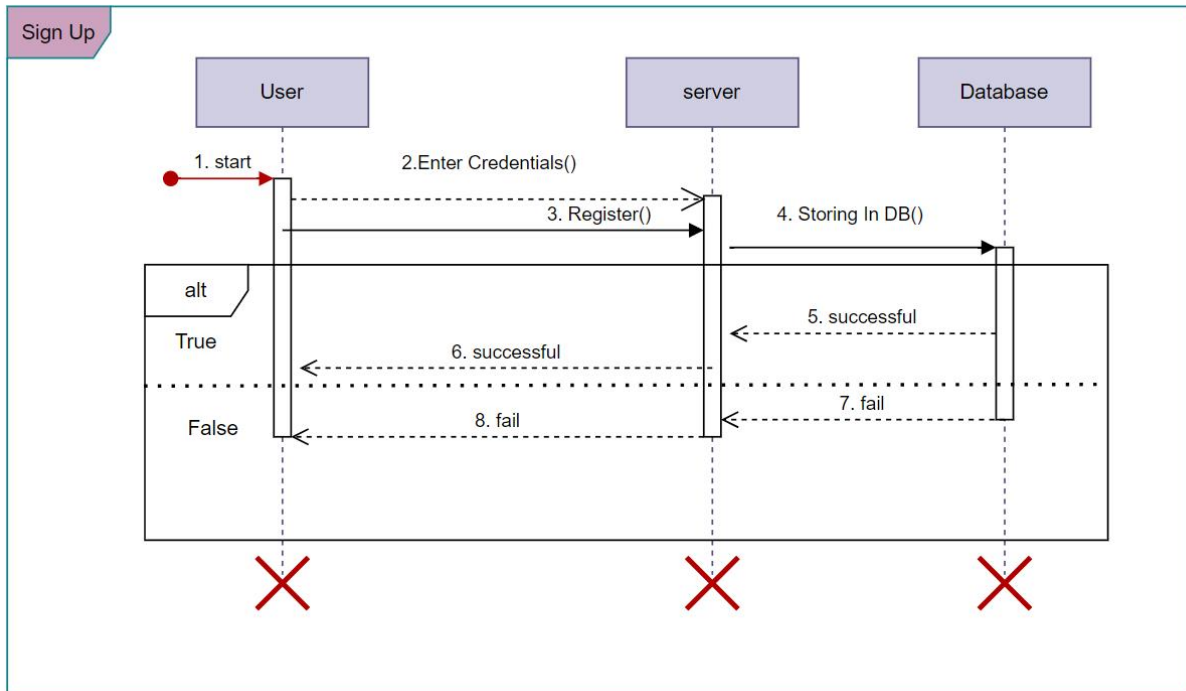


Figure 31: Sign Up Sequence Diagram

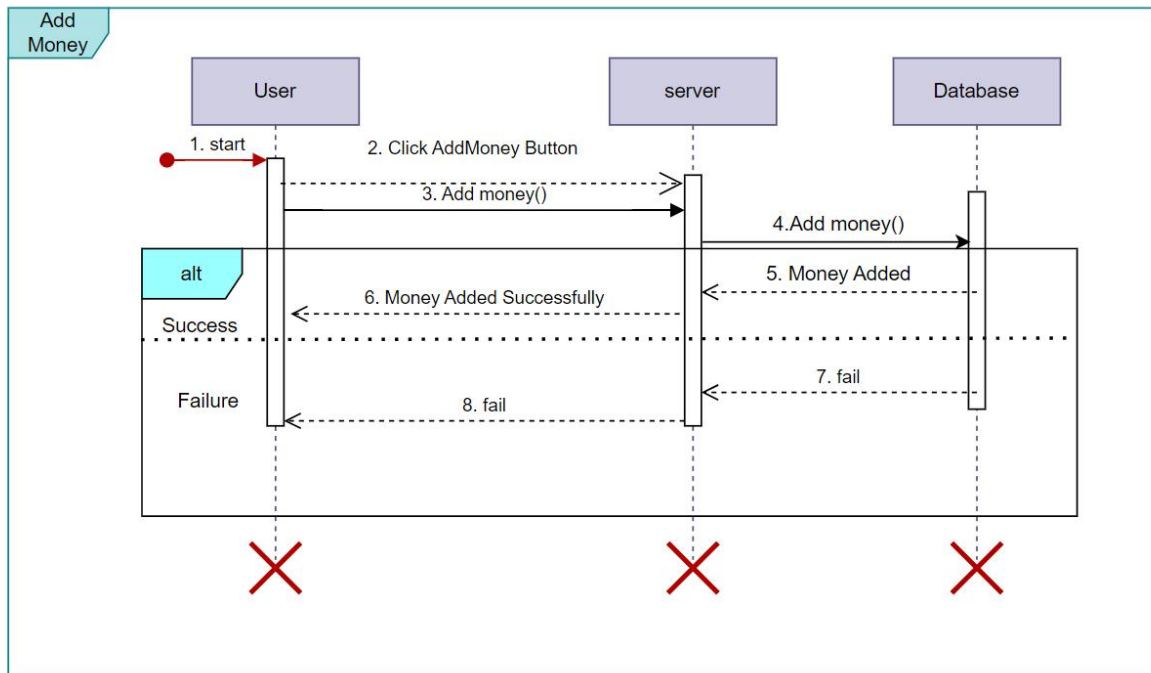


Figure 32: User Add Money

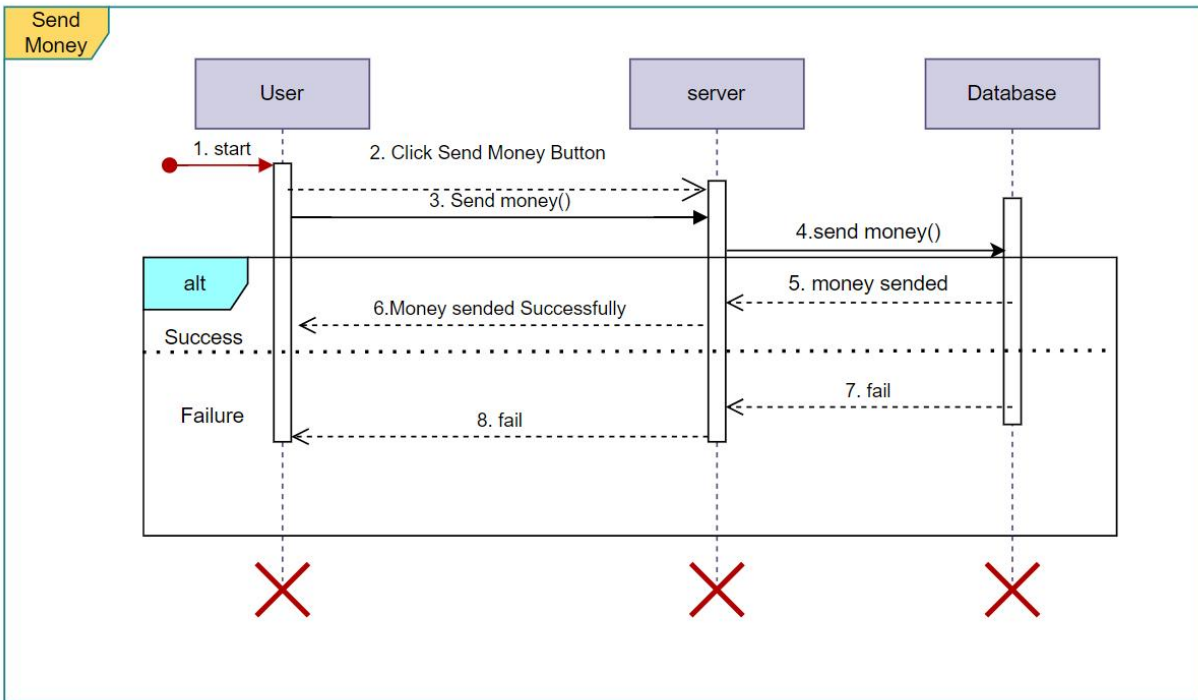


Figure 33: User Send Money

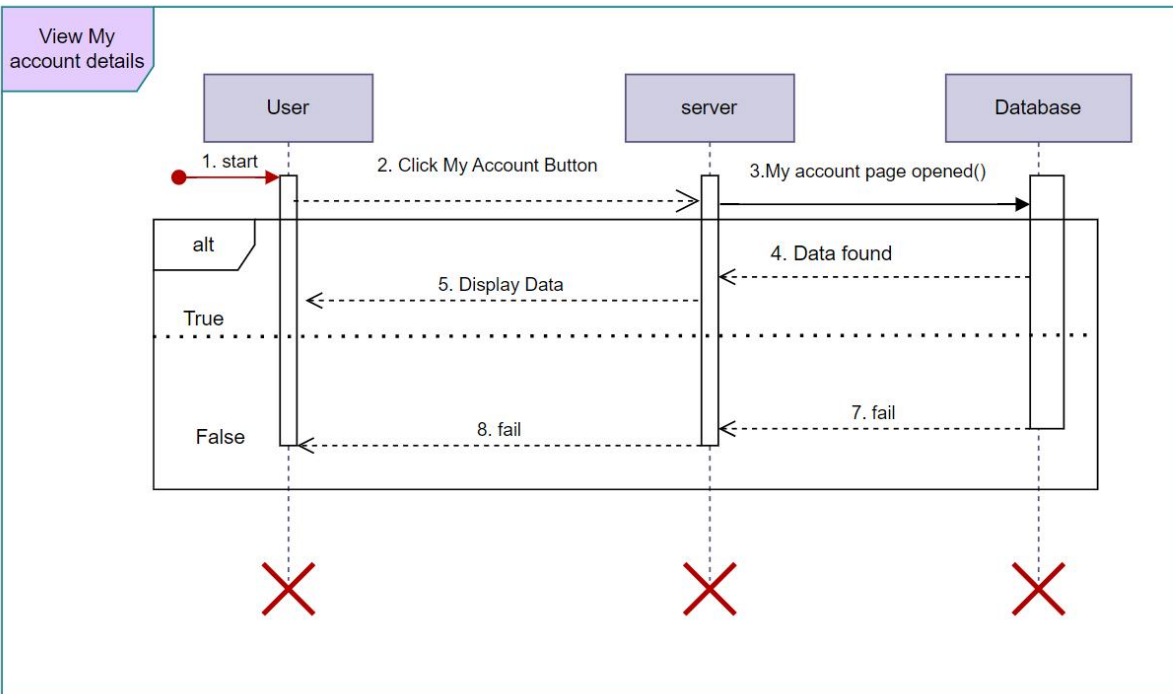


Figure 34: User View My Account

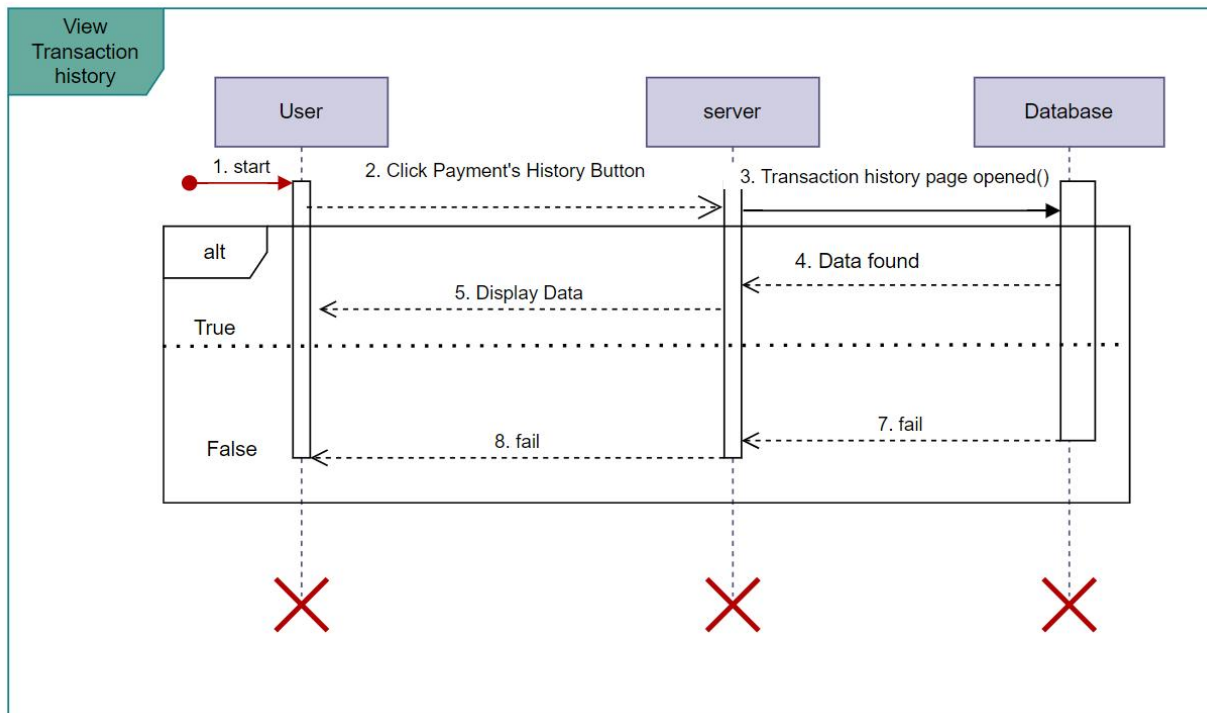


Figure 35: User View Transaction History

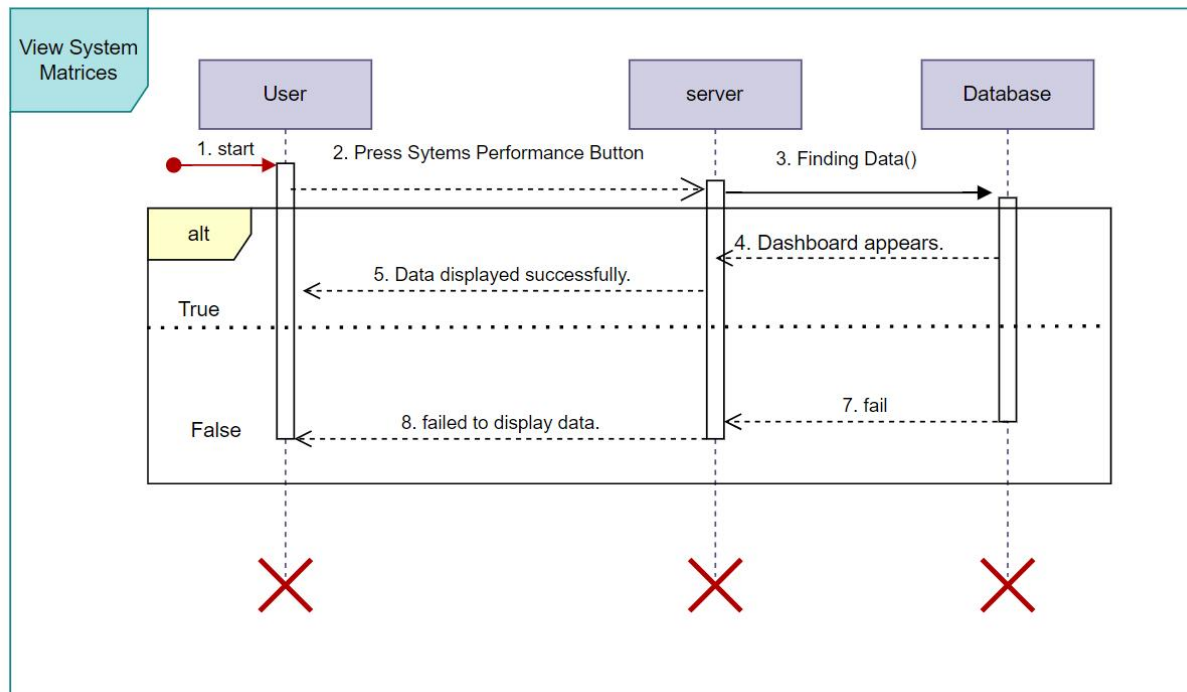


Figure 36: Admin View System Matrices

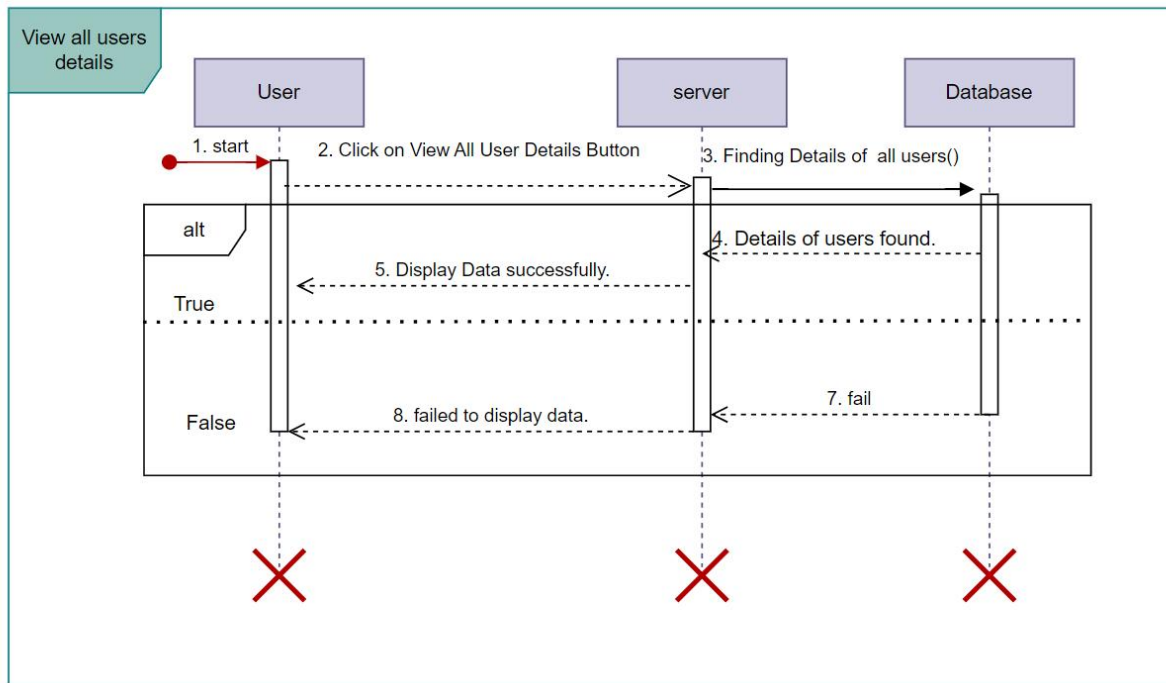


Figure 37: Admin View Bank User's Details

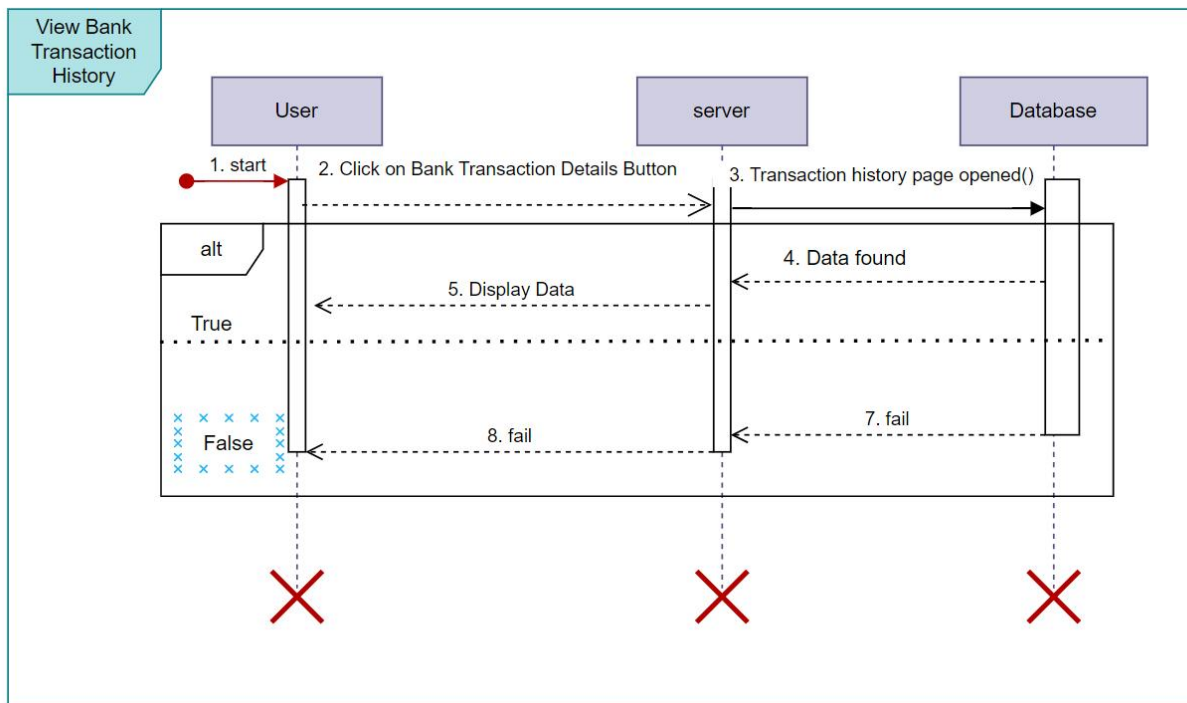


Figure 38: Admin View Bank's Transaction History

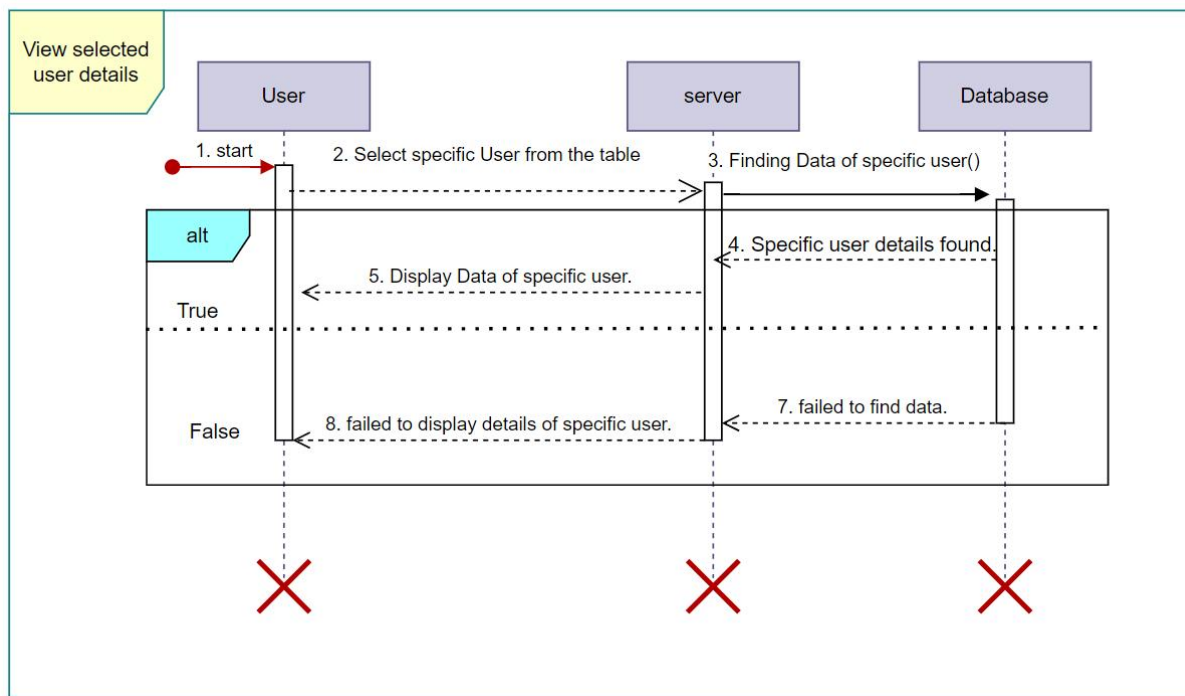


Figure 39: Admin View Selected User's Info Sequence Diagram

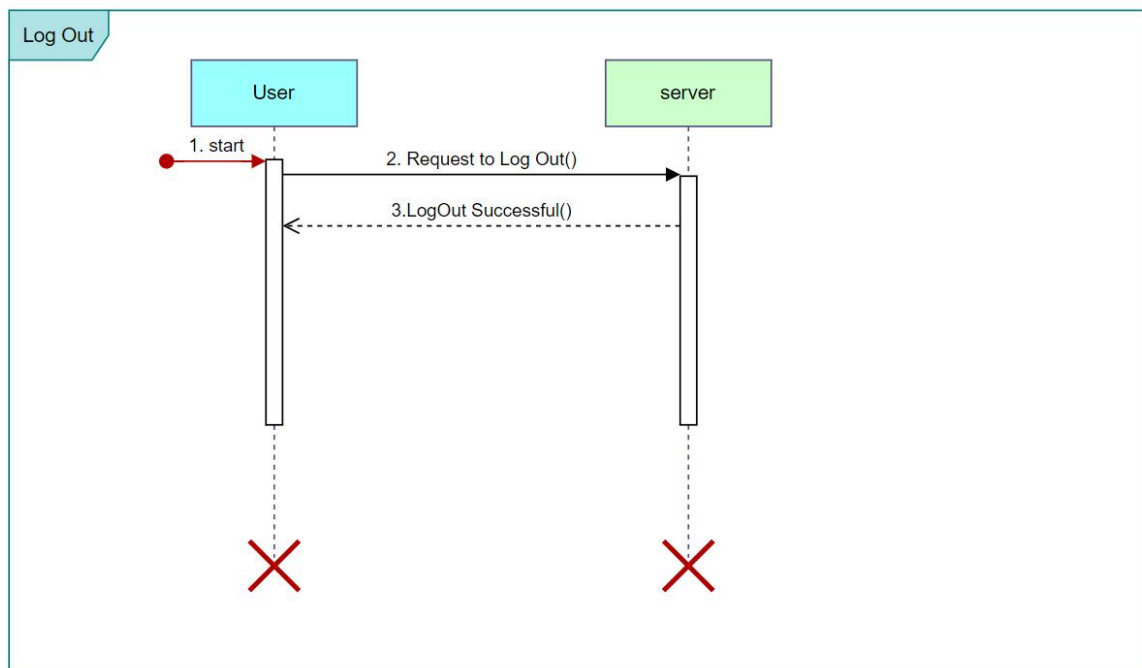


Figure 40: Log-Out Sequence Diagram

TESTING:

UNIT TESTING:

Sign-Up Screen Testing:

The module **Sign-Up Screen** is used to register new users in the system. Users must provide valid details such as their name, email, date of birth, contact number, and password to create an account successfully.

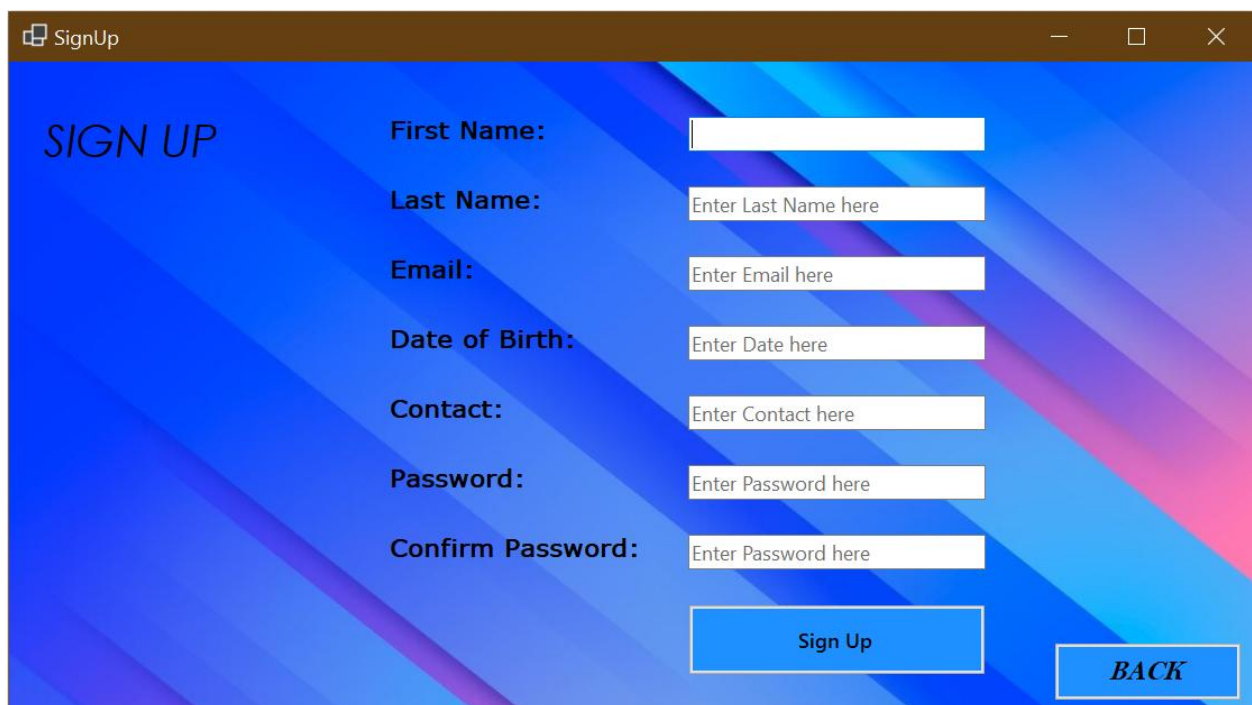
The image shows a web browser window titled "SignUp". The background is a vibrant blue and purple geometric pattern. On the left, the text "SIGN UP" is displayed in a large, white, serif font. To the right, there is a form with seven labels and corresponding text input fields: "First Name:", "Last Name:", "Email:", "Date of Birth:", "Contact:", "Password:", and "Confirm Password:". Each input field contains a placeholder text: "Enter Last Name here", "Enter Email here", "Enter Date here", "Enter Contact here", "Enter Password here", and "Enter Password here". At the bottom right of the form, there are two buttons: a blue button labeled "Sign Up" and a white button with a blue border labeled "BACK".

Figure 41: Sign-Up Form

Components of the Sign-Up Screen:

- First Name Text Field
- Last Name Text Field
- Email Text Field
- Date of Birth Text Field
- Contact Text Field
- Password Text Field
- Confirm Password Text Field

- Sign-Up Button
- Back Button

Rules:

1. All fields are mandatory and cannot be left blank.
2. Password and Confirm Password must match.
3. Only alphabets are allowed in First Name and Last Name, with no numbers or special characters, and the maximum length is 20 characters.
4. The Email must follow a valid format (e.g., abc@domain.com), and duplicate emails should not be allowed.
5. Date of Birth must be a valid date in DD-MM-YYYY or YYYY-MM-DD format, and users must be at least 18 years old.
6. The Contact field must be a valid Pakistani phone number (e.g., 0301-2345678) and can only include numbers and a dash (-).
7. Password must be at least 8 characters long, with at least one uppercase letter, one lowercase letter, one number, and one special character.
8. The Sign-Up Button should only be enabled if all fields are valid and filled. On success, the user is registered and redirected to the login page.
9. The Back Button redirects the user to the login page without saving any entered data.

Test Cases for Components:

1. First Name: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Accepted	Fail
2	HamnaNasir	Should be accepted	Accepted	Pass
3	Safwan123	Should be rejected	Rejected	Pass
4	Qasim-Mitu	Should be rejected	Rejected	Pass

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
5	@#\$\$%^	Should be rejected	Accepted	Fail
6	HamnaHamnaHamnaHamnaHamna	Should be rejected	Rejected	Pass

1. First Name: V.1.1

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	HamnaNasir	Should be accepted	Accepted	Pass
3	Safwan123	Should be rejected	Rejected	Pass
4	Qasim-Mitu	Should be rejected	Rejected	Pass
5	@#\$\$%^	Should be rejected	Rejected	Pass
6	HamnaHamnaHamnaHamnaHamna	Should be rejected	Rejected	Pass

2. Last Name: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	Safwan	Should be accepted	Accepted	Pass
3	Qasim123	Should be rejected	Rejected	Pass
4	Hamna-Nasir	Should be rejected	Rejected	Pass
5	SafwanSafwanSafwan	Should be accepted	Accepted	Pass

3. Email Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	hamna12@gmail.com	Should be accepted	Accepted	Pass
3	safwan.masood@bank.pk	Should be accepted	Accepted	Pass
4	qasim09mail.com	Should be rejected	Rejected	Pass
5	safwan@pk	Should be rejected	Rejected	Pass
6	hamna.bank@gmail.pk	Should be accepted	Accepted	Pass
7	safwan!@bank.pk	Should be rejected	Rejected	Pass

4. Date of Birth Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	15-08-2005	Should be rejected (User under 18)	Rejected	Pass
3	1990-01-01	Should be accepted	Accepted	Pass
4	01/01/1990	Should be accepted	Accepted	Pass
5	32-01-1990	Should be rejected	Rejected	Pass

5. Contact Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	0301-2345678	Should be accepted	Accepted	Pass
3	0301 2345678	Should be rejected	Rejected	Pass
4	12345-67890	Should be rejected	Rejected	Pass

6. Password Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Pa\$\$word123	Should be accepted	Accepted	Pass
2	12345678	Should be rejected	Rejected	Pass
3	password123	Should be rejected	Rejected	Pass
4	Pa@ss	Should be rejected (too short)	Rejected	Pass

7. Sign-Up Button

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Valid input in all fields	User should be registered	User registered	Pass
2	Blank input in one field	User should be asked to fill	User prompted for missing data	Pass

8. Back Button

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	Redirected to login page	Redirected	Pass

Login Screen Testing:

The module **Login Screen** is used to sign in to the system. Registered users must provide valid credentials (username and password) to access their accounts. If the user does not have an account, they can click on the "Sign Up" button to register.

The image shows a web browser window titled 'Form1' with a blue background. At the top, it says 'ONLINE BANKING APPLICATION'. Below this is a logo consisting of a stylized 'S' inside a rounded square. Under the logo, the text 'MOVING FORWARD' is displayed. To the right of the logo, there are two text input fields. The first is labeled 'User Name:' and the second is labeled 'Password:' with the placeholder text 'Enter password here'. Below these fields are two buttons: 'Login' and 'Sign Up'.

Figure 42: Login Form

Components of the Login Screen:

- Username Text Field
- Password Text Field
- Login Button
- Sign Up Button

Rules:

1. Both fields (username and password) are mandatory and cannot be left blank.
2. The username must be a valid email address following the format abc@domain.com.
3. The password must match the one provided during sign-up.
4. The Login Button should only be enabled if both fields are valid and filled.
5. Clicking the Login Button with incorrect credentials must display an error message.
6. The Sign Up Button redirects the user to the sign-up screen.

Test Cases for Components

1. Username Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	hamna12@gmail.com	Should be accepted	Accepted	Pass
3	safwan.ahmad@mail.pk	Should be accepted	Accepted	Pass
4	qasim09mail.com	Should be rejected	Rejected	Pass
5	safwan@pk	Should be rejected	Rejected	Pass
6	hamna.bank@gmail.pk	Should be accepted	Accepted	Pass
7	hamna@domain	Should be rejected	Rejected	Pass

2. Password Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	Pa\$\$word123	Should be accepted	Accepted	Pass
3	12345678	Should be rejected	Rejected	Pass
4	password123	Should be rejected	Rejected	Pass
5	WrongPass!23	Should be rejected	Rejected	Pass
6	Pa\$\$w0rd	Should be accepted	Accepted	Pass

3. Login Button

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Both fields are blank	User should be asked to fill in both fields	User prompted	Pass
2	Valid username, blank password	User should be asked to fill in the password	User prompted	Pass
3	Blank username, valid password	User should be asked to fill in the username	User prompted	Pass
4	Incorrect username & password	User should see an error message	Error message shown	Pass
5	Valid username & password	User should be logged in	User logged in successfully	Pass

4. Sign Up Button

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to sign-up page	Redirected	Pass

User Dashboard Testing:

The module **User Dashboard** serves as the main interface for a logged-in user. It provides access to key functionalities such as adding money, sending money, viewing transaction history, checking account details, and logging out.

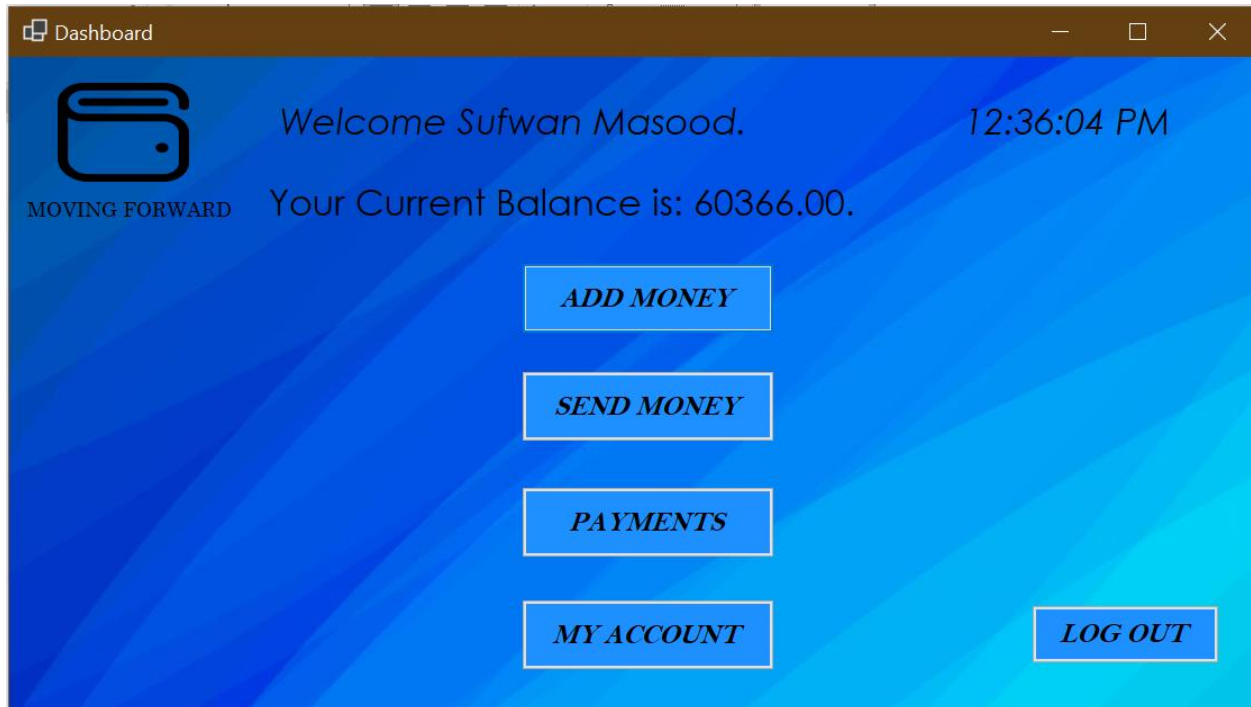


Figure 43: User Dashboard

Components of the User Dashboard:

- Add Money Button
- Send Money Button
- Payments Button (for transaction history)
- My Account Details Button
- Logout Button

Rules:

1. All buttons must be functional and lead to their respective screens.
2. Buttons should only be enabled for logged-in users.
3. Clicking the Logout Button must log the user out and redirect to the login page.

Test Cases for Components

1. Add Money Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to Add Money screen	Redirected	Pass
2	Button is disabled	User should not be able to access the screen	Access restricted	Pass

2. Send Money Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to Send Money screen	Redirected	Pass
2	Button is disabled	User should not be able to access the screen	Access restricted	Pass

3. Payments Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to Transaction History screen	Redirected	Pass
2	Button is disabled	User should not be able to access the screen	Access restricted	Pass

4. My Account Details Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to Account Details screen	Redirected	Pass
2	Button is disabled	User should not be able to access the screen	Access restricted	Pass

5. Logout Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be logged out and redirected to login page	Redirected	Pass
2	Button is disabled	User should not be able to log out	Access restricted	Pass

Add Money Page Testing:

The module **Add Money Page** allows the user to add funds to their account by providing their bank details, amount, and password. This page ensures secure and accurate transactions.

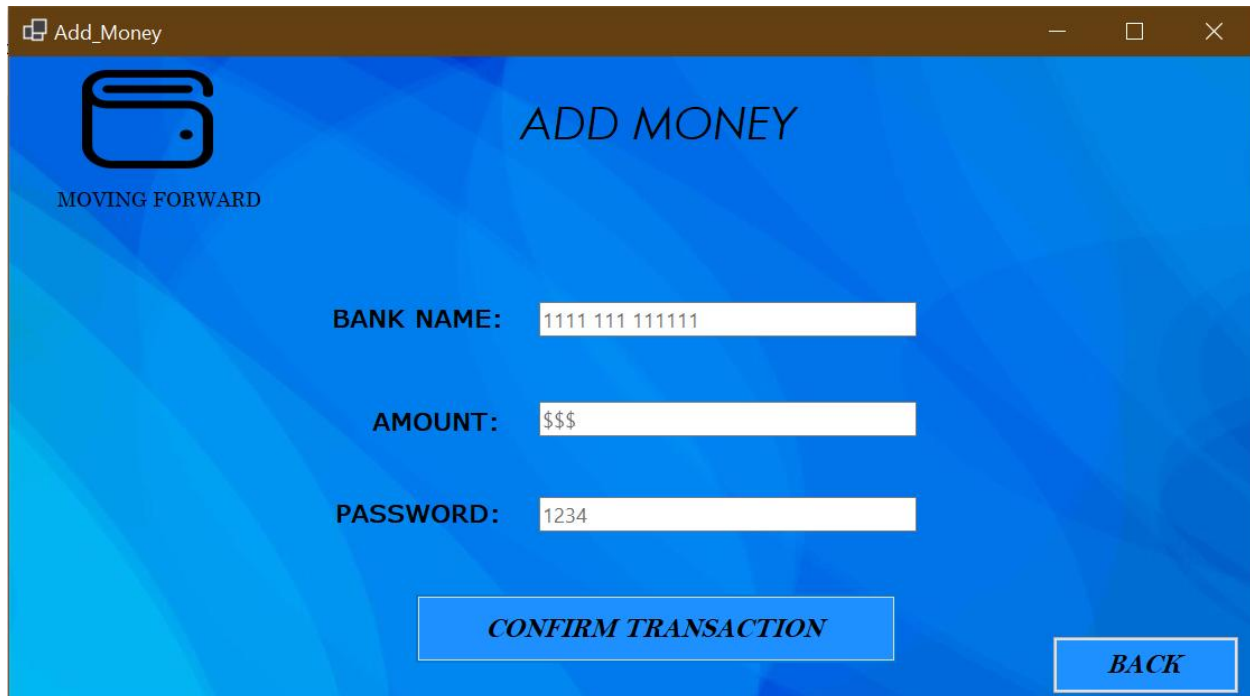
The image shows a web application window titled "Add_Money". The background is a blue gradient with a faint geometric pattern. In the top left, there is a logo consisting of a stylized 'S' inside a square, with the text "MOVING FORWARD" below it. The main heading "ADD MONEY" is displayed in a large, white, serif font. Below the heading, there are three text input fields: "BANK NAME:" with the value "1111 111 111111", "AMOUNT:" with the value "\$\$\$", and "PASSWORD:" with the value "1234". At the bottom, there are two buttons: a large blue button labeled "CONFIRM TRANSACTION" and a smaller white button labeled "BACK".

Figure 44: Add Money Form

Components of the Add Money Page:

- Bank Name Text Field
- Amount Text Field
- Password Text Field
- Confirm Transaction Button
- Back Button

Rules:

1. All fields (Bank Name, Amount, and Password) are mandatory and cannot be left blank.
2. Bank Name must be selected from the predefined list of valid banks.

3. Amount must be a positive number, greater than zero, and within the transaction limit (e.g., not exceeding PKR 1,000,000).
4. Password must match the user's account password.
5. Clicking the Confirm Transaction Button must validate all inputs, complete the transaction, and update the account balance.
6. Clicking the Back Button must redirect the user to the User Dashboard without saving any entered data.

Test Cases for Components

1. Bank Name Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	Meezan Bank	Should be accepted	Accepted	Pass
3	Bank Alfalah	Should be accepted	Accepted	Pass
4	HBL	Should be accepted	Accepted	Pass
5	Unknown Bank	Should be rejected	Rejected	Pass

2. Amount Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	-500	Should be rejected	Rejected	Pass
3	0	Should be rejected	Rejected	Pass
4	100	Should be accepted	Accepted	Pass
5	1000000	Should be accepted	Accepted	Pass
6	1000001	Should be rejected (exceeds limit)	Rejected	Pass

3. Password Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	WrongPassword123	Should be rejected	Rejected	Pass
3	CorrectPassword!23	Should be accepted	Accepted	Pass
4	12345	Should be rejected (too short)	Rejected	Pass

4. Confirm Transaction Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Valid Bank, Amount, and Password	Transaction should be completed successfully	Transaction completed	Pass
2	Blank Bank Name	User should be prompted to enter the Bank Name	Prompted	Pass
3	Blank Amount	User should be prompted to enter the Amount	Prompted	Pass
4	Blank Password	User should be prompted to enter the Password	Prompted	Pass
5	Invalid Password	Transaction should be rejected	Transaction rejected	Pass
6	Amount exceeds limit (e.g., PKR 1,000,000)	Transaction should be rejected	Transaction rejected	Pass

5. Back Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to User Dashboard	Redirected	Pass

Send Money Page Testing:

The **Send Money Page** allows users to transfer money to another account by providing the recipient's IBAN, the amount to be transferred, and their password.

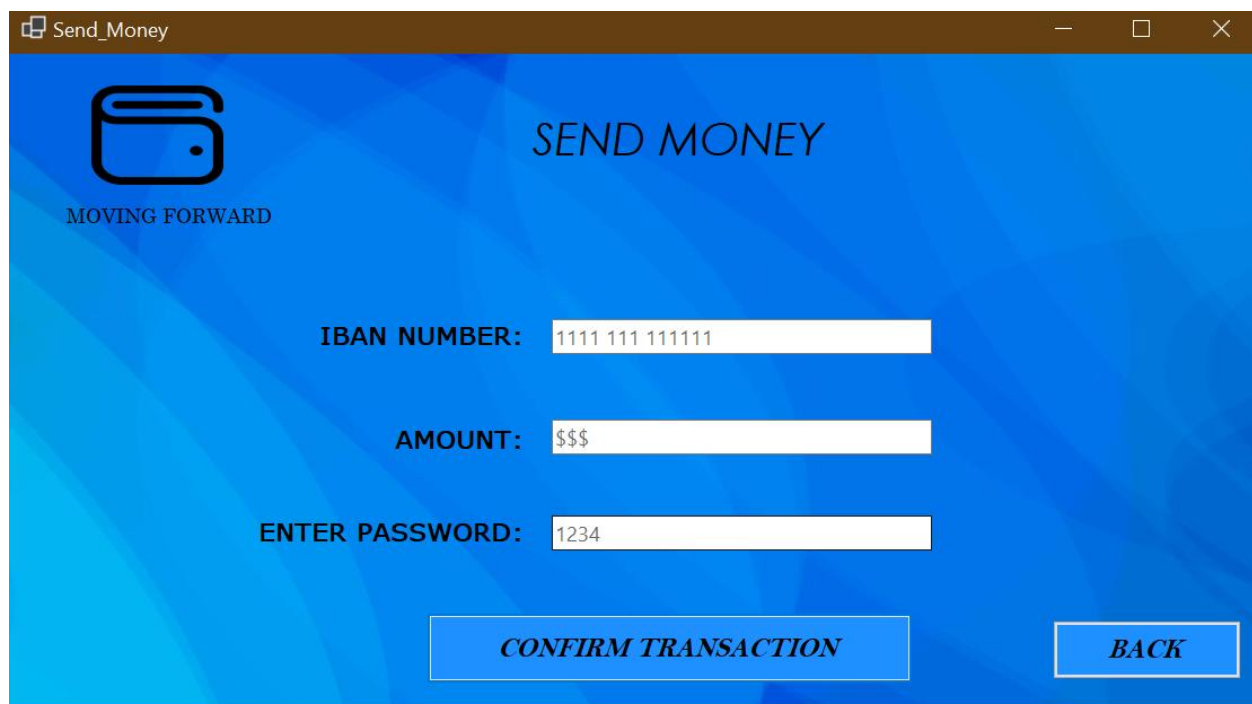
The image shows a web browser window titled "Send_Money". The page has a blue background with a subtle geometric pattern. In the top left, there is a logo consisting of a stylized 'S' and 'M' inside a square, with the text "MOVING FORWARD" below it. To the right of the logo, the text "SEND MONEY" is displayed in a large, white, serif font. Below this, there are three input fields: "IBAN NUMBER:" with the value "1111 111 111111", "AMOUNT:" with the value "\$\$\$", and "ENTER PASSWORD:" with the value "1234". At the bottom of the form, there are two buttons: "CONFIRM TRANSACTION" and "BACK".

Figure 45: Send Money Form

Components of the Send Money Page:

- IBAN Text Field
- Amount Text Field

- Password Text Field
- Confirm Transaction Button
- Back Button

Rules:

1. All fields (IBAN, Amount, Password) are mandatory and cannot be left blank.
2. The IBAN must follow a valid format (e.g., PKBANK123456).
3. Amount must be positive, greater than zero, and not exceed the sender's current balance.
4. Password must match the user's account password.
5. Clicking the Confirm Transaction Button must validate all inputs, perform the transaction, and update the database.
6. Clicking the Back Button must redirect the user to the User Dashboard without saving any entered data.

Test Cases for Components

1. IBAN Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	PKBANK123456	Should be accepted	Accepted	Pass
3	1234567890	Should be rejected	Rejected	Pass
4	PKBANK-123456	Should be rejected	Rejected	Pass
5	PKBANK0987654321	Should be accepted	Accepted	Pass

2. Amount Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	-100	Should be rejected	Rejected	Pass
3	0	Should be rejected	Rejected	Pass
4	500	Should be accepted	Accepted	Pass
5	10000000	Should be rejected (exceeds limit)	Rejected	Pass

3. Password Text Field: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	(Blank Input)	Should be rejected	Rejected	Pass
2	CorrectPassword!23	Should be accepted	Accepted	Pass
3	WrongPassword123	Should be rejected	Rejected	Pass

4. Confirm Transaction Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Valid IBAN, Amount, and Password	Transaction should be completed successfully	Transaction completed	Pass
2	Blank IBAN	User should be prompted to enter the IBAN	Prompted	Pass
3	Invalid Amount	User should be prompted to enter a valid amount	Prompted	Pass

5. Back Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to User Dashboard	Redirected	Pass

Payments Page Testing:

The **Payments Page** shows the user's transaction history, divided into deposits and withdrawals, using two separate tables.

The screenshot shows a web application window titled "Payments". The main heading is "TRANSACTION HISTORY" with a wallet icon and the text "MOVING FORWARD". Below this, there are two tables: "DEPOSITS" and "WITHDRAWS".

DEPOSITS Table:

	Sender IBAN	
	easypaisa	1
	jazzcash	1
	easypaisa	1
	jazzcash	1
▶	jazzcash	1
	easypaisa	1

WITHDRAWS Table:

	Recipient IBAN	
	PKBANK000002	
	PKBANK000002	
▶	PKBANK000003	
	PKBANK000002	
	PKBANK000003	
	PKBANK000002	

A blue "BACK" button is located at the bottom right of the page.

Figure 46: User Payment's History Form

Components of the Payments Page:

- Deposits Table (Transaction ID, Amount, Sender IBAN, Date)
- Withdrawals Table (Transaction ID, Amount, Recipient IBAN, Date)
- Back Button

Rules:

1. Both tables must display the correct transaction details for the logged-in user.
 2. The Deposits Table must show valid data (Transaction ID, Amount, Sender IBAN, and Date).
 3. The Withdrawals Table must show valid data (Transaction ID, Amount, Recipient IBAN, and Date).
 4. Clicking the Back Button must redirect the user to the User Dashboard.
-

Test Cases for Components

1. Deposits Table: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Correct deposit details	Should display accurate data	Accurate data displayed	Pass
2	Missing transaction ID	Should not display incomplete rows	Incomplete rows not displayed	Pass
3	Incorrect Sender IBAN	Should not display invalid IBANs	Invalid IBANs not displayed	Pass

2. Withdrawals Table: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Correct withdrawal details	Should display accurate data	Accurate data displayed	Pass
2	Missing transaction ID	Should not display incomplete rows	Incomplete rows not displayed	Pass

3. Back Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to User Dashboard	Redirected	Pass

My Account Info Page Testing:

The **My Account Info Page** displays the logged-in user's personal information, including balance, contact, IBAN, and other details provided during sign-up.

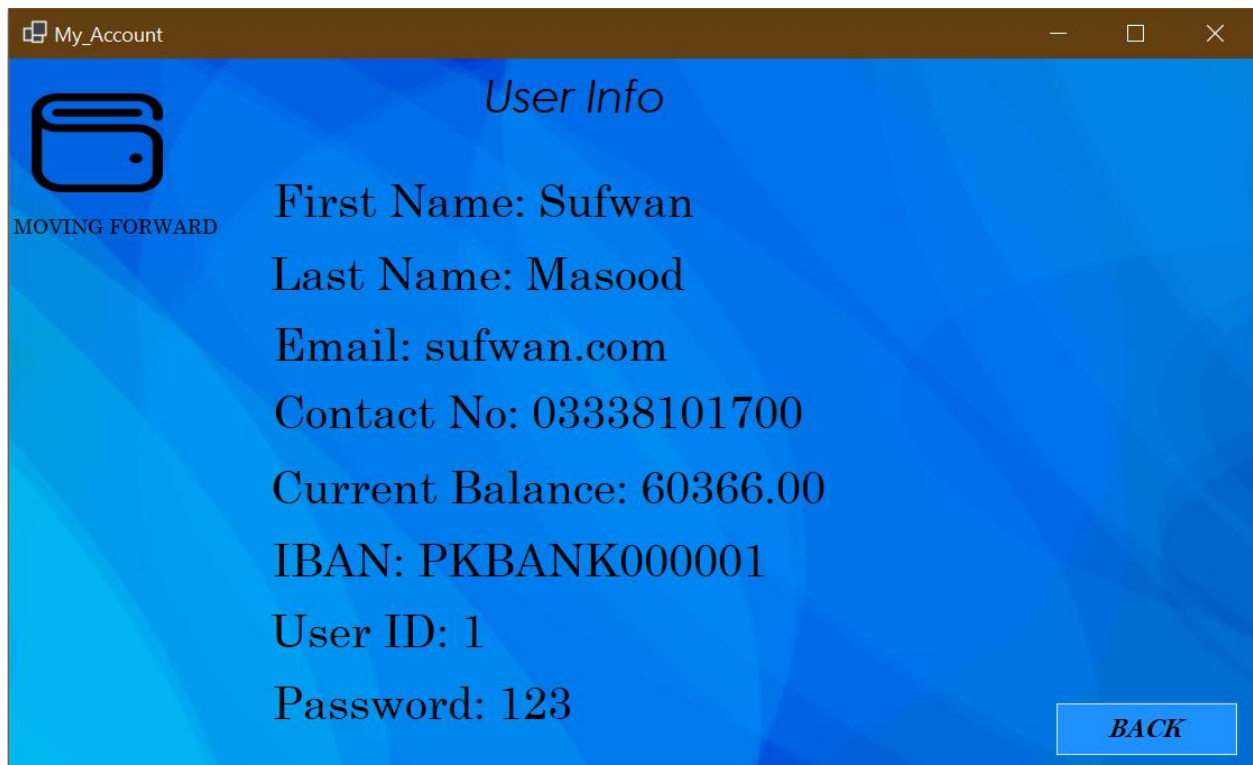


Figure 47: User's Info Page

Components of the My Account Info Page:

- User Details Section (First Name, Last Name, IBAN, Contact, etc.)
- Back Button

Rules:

1. All user details must be correctly fetched and displayed from the database.
 2. The balance displayed must match the current account balance.
 3. Clicking the Back Button must redirect the user to the User Dashboard.
-

Test Cases for Components

1. User Details Section: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Correct user details	All details should be displayed correctly	Details displayed correctly	Pass
2	Missing or incomplete details	Should not display incomplete information	Incomplete info not displayed	Pass

2. Back Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	User should be redirected to User Dashboard	Redirected	Pass

Admin Panel Testing:

The **Admin Panel** provides a comprehensive dashboard for the administrator to monitor system statistics, view user details, and access other administrative functionalities.

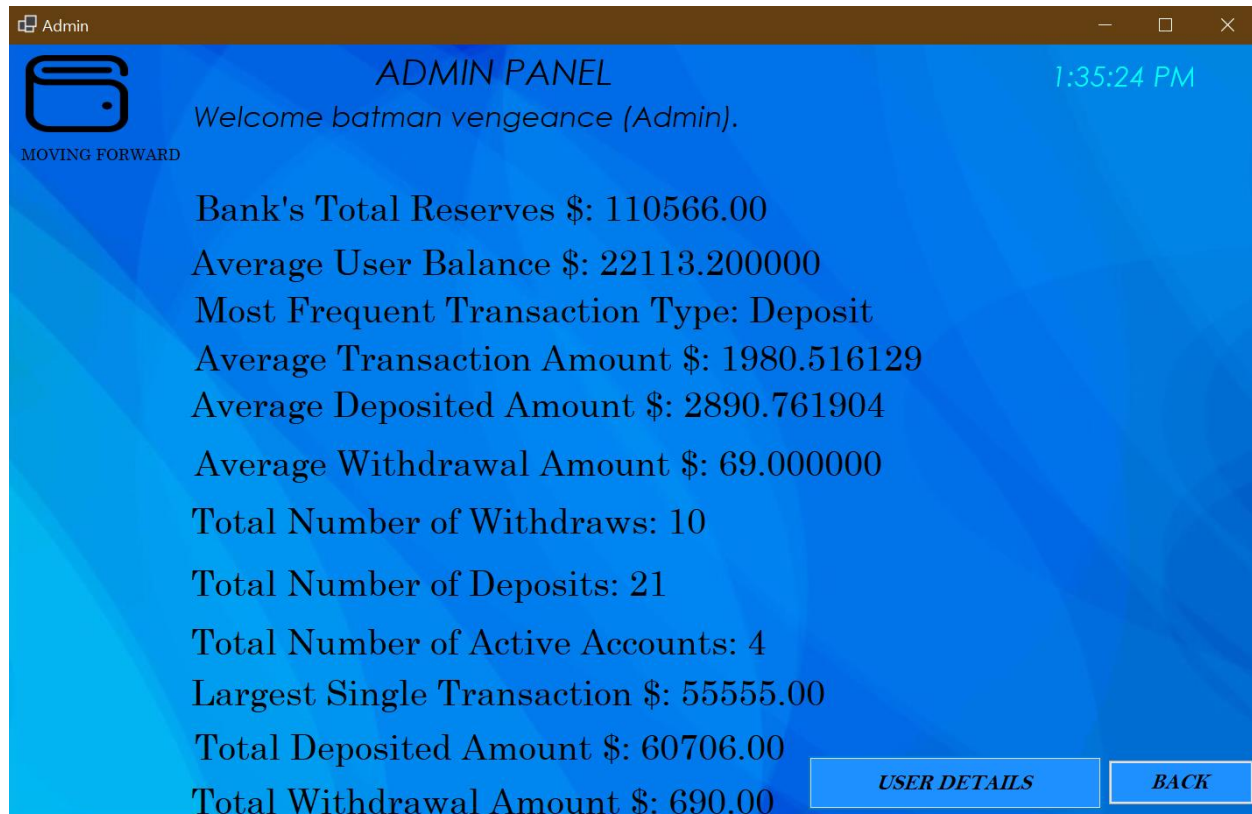


Figure 48: Admin Panel Form

Components of the Admin Panel:

- Welcome Admin Message
 - Clock Display
 - Back Button (leads to the Sign-In page)
 - User Details Button (leads to the User Details page)
 - Statistics Section
-

Rules:

- All statistics must display accurate and updated values fetched from the database.
 - The Back Button must log out the admin and redirect to the Sign-In page.
 - The User Details Button must redirect to the User Details page.
 - The Clock must display the correct real-time system time.
-

Test Cases for Components:

1. Statistics Section: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Total Bank Reserves	Correct value should be displayed	Correct value displayed	Pass
2	Average User Balance	Correct value should be displayed	Correct value displayed	Pass
3	Most Frequent Transaction Type	Correct value should be displayed	Correct value displayed	Pass
4	Total Deposits	Correct value should be displayed	Correct value displayed	Pass
5	Total Withdrawals	Correct value should be displayed	Correct value displayed	Pass
6	Largest Transaction Amount	Correct value should be displayed	Correct value displayed	Pass

2. Back Button: V.1.0

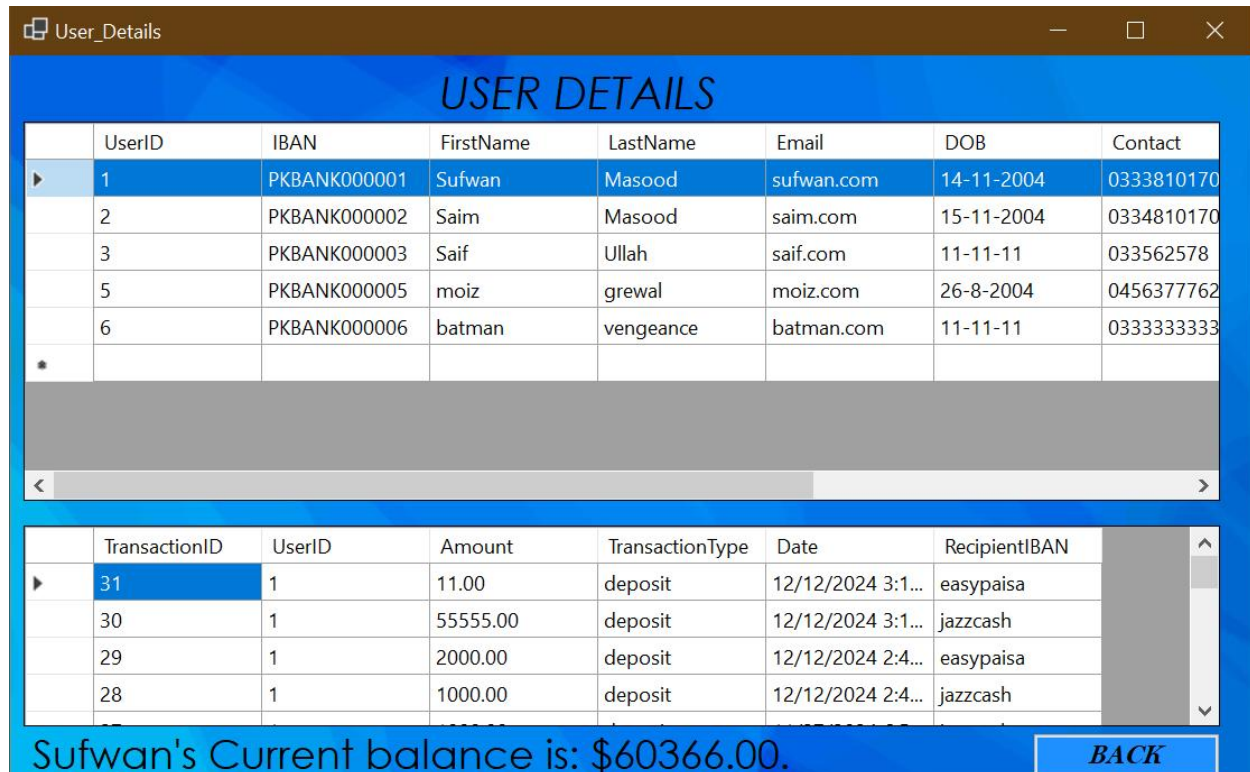
Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	Admin should be redirected to Sign-In page	Redirected	Pass

3. User Details Button: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Button is clicked	Admin should be redirected to User Details page	Redirected	Pass

User Details Page Testing:

The **User Details Page** provides an overview of all users in the system, allowing the admin to select a user for more detailed transaction history.



The screenshot shows a web application window titled "User_Details". The main heading is "USER DETAILS". Below it is a table with columns: UserID, IBAN, FirstName, LastName, Email, DOB, and Contact. The table contains six rows of user data. Below this table is a horizontal scrollbar. Underneath the scrollbar is another table with columns: TransactionID, UserID, Amount, TransactionType, Date, and RecipientIBAN. This table shows four transaction records for the selected user. At the bottom of the page, a blue banner displays the text "Sufwan's Current balance is: \$60366.00." and a "BACK" button.

UserID	IBAN	FirstName	LastName	Email	DOB	Contact
1	PKBANK000001	Sufwan	Masood	sufwan.com	14-11-2004	0333810170
2	PKBANK000002	Saim	Masood	saim.com	15-11-2004	0334810170
3	PKBANK000003	Saif	Ullah	saif.com	11-11-11	033562578
5	PKBANK000005	moiz	grewal	moiz.com	26-8-2004	0456377762
6	PKBANK000006	batman	vengeance	batman.com	11-11-11	0333333333

TransactionID	UserID	Amount	TransactionType	Date	RecipientIBAN
31	1	11.00	deposit	12/12/2024 3:1...	easypaisa
30	1	55555.00	deposit	12/12/2024 3:1...	jazzcash
29	1	2000.00	deposit	12/12/2024 2:4...	easypaisa
28	1	1000.00	deposit	12/12/2024 2:4...	jazzcash

Sufwan's Current balance is: \$60366.00. [BACK](#)

Figure 49: Admin View User's Details

Components of the User Details Page:

- Users Table (User ID, First Name, Last Name, Email, DOB, Contact, Password, Balance)
- Transaction Details Table (appears on double-click; Transaction ID, Date, Amount, Recipient IBAN)

- Prompt Message (prompts admin to double-click a user for details)

Rules:

1. The Users Table must display accurate and updated details of all users.
2. Double-clicking a user must open the Transaction Details Table with that user's specific transaction history.
3. The Prompt Message must be displayed by default.
4. If no transactions exist for the selected user, the Transaction Details Table should display "No transactions found."

Test Cases for Components

1. Users Table: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	All user details fetched	Table should display correct user details	Correct details displayed	Pass
2	User with empty fields	Table should not display incomplete user details	Incomplete rows not displayed	Pass

2. Double-Click User (Transaction Details Table): V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	User has transactions	Correct transaction history should be displayed	History displayed	Pass
2	User has no transactions	Table should display "No transactions found"	Message displayed	Pass
3	Invalid user selected (does not exist)	Table should not display any transaction history	Table not displayed	Pass

3. Prompt Message: V.1.0

Sr. No	Test Data	Expected Result	Actual Result	Pass/Fail
1	Page is loaded	Prompt message should be displayed	Prompt message displayed	Pass

INTEGRATION TESTING:

After each unit is thoroughly tested, it is integrated with other units to create modules or components that are designed to perform specific tasks or activities. These are then tested as group through integration testing to ensure whole segments of an application behave as expected (i.e. the interactions between units are seamless). These tests are often framed by user scenarios, such as logging into an application or opening files. Integrated modules are usually comprised of a combination of automated functional and manual tests.

Add Money Integration Testing:

User must be able to add money to their account and see updated balance.

Table 1.2: Add Money Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	Add money to account	Money should be added, balance updated, and logged in transaction history	Transaction completed as expected.	Pass
2	Add money error	Invalid payment details should show an error message	Error shown prompting user to try again.	Pass

Send Money Integration Testing:

User must be able to send money to other accounts.

Table 1.3: Send Money Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	Successful Send Money Transaction	Funds should be transferred and logged in transaction history	Transaction completed	Pass
2	Send Money Error Handling	Invalid recipient details or insufficient balance should show an error message	Error shown prompting user to try again.	Pass

View Payments Integration Testing:

User must be able to view transaction history.

Table 1.4: View Payments Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	View deposits	User must see accurate details of deposits	Deposits shown	Pass
2	View withdrawals	User must see accurate details of withdrawals	Withdrawals shown	Pass

View My Account Info Integration Testing:

User must be able to view their account information.

Table 1.5: View My Account Info Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	View personal information	User must see their details such as name, contact, and IBAN	Details shown	Pass
2	View balance	User must see their current account balance	Balance correct	Pass

Login and Sign-Up Integration Testing:

User must be able to create and log in to their account.

Table 1.1: Login and Sign-Up Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	User sign-up and login	User must be able to sign up and log in successfully and redirected to user dashboard	Signed In and jumped to user panel	Pass
2	Admin login	Admin must be able to log in and access the admin panel	Able to log in	Pass

3	User not exist	User must not be able to log in if not registered	Cannot sign in	Pass
---	----------------	---	----------------	------

Logout Integration Testing:

User must be able to log out of their account.

Table 1.9: Logout Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	User logout functionality	User must be logged out and redirected to the login page	Logout successful	Pass
2	Admin logout functionality	Admin must be logged out and redirected to the admin login page	Logout successful	Pass

Admin - View System Metrics Integration Testing:

Admin must be able to view overall system metrics.

Table 1.6: Admin - View System Metrics Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	View total bank reserves	Admin must see correct total reserves	Reserves correctly shown	Pass
2	View average user balance	Admin must see correct average user balance	Balance shown is correct	Pass

Admin - View Active Users Integration Testing:

Admin must be able to view details of all active users.

Table 1.7: Admin - View Active Users Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	View all active users	Admin must see accurate details of active users	Users info is correct	Pass

Admin - View Selected User Transaction Details Integration Testing:

Admin must be able to view transaction history of selected users.

Table 1.8: Admin - View Selected User Transaction Details Integration Test Result

Test Case	Test Case Description	Expected Result	Actual Result	Pass/Fail
1	View user transactions	Admin must see correct transaction history of the selected user	Transactions are shown as expected.	Pass

SYSTEM TESTING:

Sign-In Testing:

Test Case ID	BU_001	Test Case Description	Test the Login Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	1.1

QA Tester's Log	Review comments from Hamna incorporate in version 1.1
-----------------	---

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	A valid registered account should exist in the system.
3	Access to the banking system application
4	

S #	Test Data
1	Email: Hamna_9066@gmail.com
2	Pass = 1234to?
3	
4	

Test Scenario	Verify on entering valid Email and password, so the customer can login.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As <u>Expected</u>	Pass
2	Enter Email & Password	Credential should be entered	As <u>Expected</u>	Pass
3	Click on Login Button	User is logged in successfully	As <u>Expected</u>	Pass

Post Condition	The user is logged in and redirected to the dashboard to access their account features.
----------------	---

Sign-Up Testing:

Test Case ID	BU_001	Test Case Description	Test the Sign-Up Functionality in Banking		
Created By	Safwan	Reviewed By	Hamna	Version	2.1

QA Tester's Log	Review comments from Hamna incorporate in version 2.1
-----------------	---

Tester's Name	Saim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	Ensure the user does not already have an account in the system.
3	Access to the banking system application
4	

S #	Test Data
1	Email: Sufwan@gmail.com
2	Password = Batman
3	Full Name = Sufwan Masood
4	Contact = 032326731
5	Address = House 177D, GT, Ph.3, Gujranwala.
6	Date Of Birth = 14-11-2004
7	Confirm Password = Batman

Test Scenario	Verify a new user can successfully create an account.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Click the "Sign-Up" button	The sign-up page should load	As Expected	Pass
3	Enter user details	Details should be entered successfully	As Expected	Pass
4	Click the "Create Account" button	Account is created and user is redirected to login	As Expected	Pass

Post Condition	The user is registered and can log in using their credentials.
----------------	--

Add Money Testing:

Test Case ID	BU_001	Test Case Description	Test the Add Money Functionality in Banking		
Created By	Safwan	Reviewed By	Hamna	Version	2.1

QA Tester's Log	Review comments from Hamna incorporate in version 2.1
-----------------	---

Tester's Name	Tayyab	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	--------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	User must be logged in to the system.
3	Access to the banking system application
4	Ensure the user has a valid bank account linked

S #	Test Data
1	Bank Name: JazzCash Bank
2	Amount: 5000 PKR
3	Password: Password@123

Test Scenario	Verify the user can successfully add money to their account.
---------------	--

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	User is logged in successfully	As Expected	Pass
4	Navigate to the "Add Money" page	Add Money page should load successfully	As Expected	Pass
5	Enter bank details and amount	Details should be entered successfully	As Expected	Pass
6	Click the "Confirm Transaction" button	Money is added to the account balance	As Expected	Pass

Post Condition	The user's account balance is updated, and the transaction is logged. _____
----------------	---

Send Money Testing:

Test Case ID	BU_003	Test Case Description	Test the Send Money Functionality in Banking		
Created By	Safwan	Reviewed By	Hamna	Version	3.1

QA Tester's Log	Review comments from Hamna incorporate in version 3.1
-----------------	---

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	User must be logged in to the system.
3	Access to the banking system application
4	Ensure the user has sufficient account balance.

S #	Test Data
1	Recipient IBAN: PKBANK123456
2	Amount: 3000 PKR
3	Password: Password@123

Test Scenario	Verify the user can successfully send money to another account.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	User is logged in successfully	As Expected	Pass
4	Navigate to the "Send Money" page	Send Money page should load successfully	As Expected	Pass
5	Enter bank details and amount	Details should be entered successfully	As Expected	Pass
6	Click the "Confirm Transaction" button	Money is transferred successfully	As Expected	Pass

Post Condition	The transaction is recorded, and the sender's balance is updated.
----------------	---

View Transaction History Testing:

Test Case ID	BU_004	Test Case Description	Test the View Payments Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	4.1

QA Tester's Log	Review comments from Hamna incorporate in version 4.1
-----------------	---

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	User must be logged in to the system.
3	Access to the banking system application
4	Ensure the user has sufficient transaction history to show.

S #	Test Data
1	Email: Sufwan@gmail.com
2	Password: Password@123

Test Scenario	Verify the user can view transaction history.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	User is logged in successfully	As Expected	Pass
4	Navigate to the "View Transaction" page	Transaction history is displayed accurately	As Expected	Pass

Post Condition	The user views all transaction history, including deposits and withdrawals
----------------	--

View My Account Testing:

Test Case ID	BU_005	Test Case Description	Test the My Account Details in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	5.1

QA Tester's Log	Review comments from Hamna incorporate in version 5.1
-----------------	---

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	User must be logged in to the system.
3	Access to the banking system application
4	Ensure the user has account info to display.

S #	Test Data
1	Email: Sufwan@gmail.com
2	Password: Password@123

Test Scenario	Verify the user can view his/her account details.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	User is logged in successfully	As Expected	Pass
4	Navigate to the "My Account" page	My account Information is displayed successfully on the screen	As Expected	Pass

Post Condition	The user views his personal account information.
----------------	--

Logout Testing:

Test Case ID	BU_006	Test Case Description	Test the Log Out Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	6.1

QA Tester's Log	Review comments from Hamna incorporate in version 6.1
-----------------	---

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	User must be logged in to the system.
3	Access to the banking system application

S #	Test Data
1	Email: Sufwan@gmail.com
2	Password: Password@123

Test Scenario	Verify the user can view his/her account details.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Logout Button	User is logged out successfully	As Expected	Pass

Post Condition	The user is logged out of the system and cannot access the dashboard without logging back in.
----------------	---

Admin View Bank's Statistics:

Test Case ID	BU_010	Test Case Description	Test the Admin View Bank Matrices Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	10.1

QA Tester's Log	Review comments from Hamna incorporate in version 10.1
-----------------	--

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	Admin must be logged in to the system.
3	Access to the banking system application
4	Ensure the Bank has statistical data to display

S #	Test Data
1	Email: Admin@gmail.com
2	Password: admin@123

Test Scenario	Verify the admin can view statistical data of the bank.
---------------	---

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	Admin is logged in successfully	As Expected	Pass
4	Navigate to the "View Bank's Matrices" page	Statistical data is displayed accurately	As Expected	Pass

Post Condition	The Admin View all the bank's large scale data on screen.
----------------	---

View Active User's:

Test Case ID	BU_010	Test Case Description	Test the Admin View Active Users Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	10.1

QA Tester's Log	Review comments from Hamna incorporate in version 10.1
-----------------	--

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	Admin must be logged in to the system.
3	Access to the banking system application
4	Ensure the Bank has active user's to display their general info.

S #	Test Data
1	Email: Admin@gmail.com
2	Password: admin@123

Test Scenario	Verify the admin can view the general records of active users.
---------------	--

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	Admin is logged in successfully	As Expected	Pass
4	Click on the "View Active Users" page	General records of active users is displayed accurately	As Expected	Pass

Post Condition	The Admin View all the bank's active user data on screen.
----------------	---

View Selected User's details:

Test Case ID	BU_012	Test Case Description	Test the Admin View Selected User's Info Functionality in Banking Application		
Created By	Safwan	Reviewed By	Hamna	Version	12.1

QA Tester's Log	Review comments from Hamna incorporate in version 12.1
-----------------	--

Tester's Name	Qasim	Date Tested	December 21, 2024	Test Case (Pass/Fail/Not Executed)	Pass
---------------	-------	-------------	-------------------	------------------------------------	------

S #	Prerequisites:
1	Have an internet connection.
2	Admin must be logged in to the system.
3	Access to the banking system application
4	Ensure the selected user has valid records to display

S #	Test Data
1	Email: Admin@gmail.com
2	Password: admin@123

Test Scenario	Verify the admin can view the complete records of selected users.
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Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open Banking Application	The desktop application should be opened	As Expected	Pass
2	Enter Email & Password	Credential should be entered	As Expected	Pass
3	Click on Login Button	Admin is logged in successfully	As Expected	Pass
4	Click on the "Active User's" page	General records of active users is displayed accurately	As Expected	Pass
5	Double Click on the user from the active user's list	Complete Information of the selected user is shown	As Expected	Pass

Post Condition	The Admin View all the details of the selected user on the screen.
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